

02

Damodaran, the Important Parts

A revision brain-dump for someone coming back to valuation after a long gap

Understand the project.

Know what deserves your attention.

COLORFUL READING EDITION

The complete original text, with reading maps, priority labels and selective highlights. Start with the map on the next page.

Your reading map

The valuation ideas behind the machine

Start with the links below. Then read the full text at your own pace.

- **Chapter 1 / Story and the 3P test** - Separate price, value, and plausible assumptions.
- **Chapter 2 / The base year** - Know where the starting numbers come from.
- **Chapter 3 / The nine-station DCF** - Follow Almarai from revenue to value per share.
- **Chapter 4 / Discount rates** - Match risk, currency, and dated inputs.
- **Chapter 5 / Growth and terminal value** - Understand reinvestment, defaults, and ranges.
- **Chapter 6 / What the market prices in** - Keep pricing and valuation distinct.
- **Chapter 7 / India and the checklist** - See which inputs change across countries.

Keep for later. Read a chapter at a time. Its highlighted closing takeaways are your checkpoint; source lists and switch details are references.

The four labels

Label	How to use it
READ FIRST	Blue. Essential to understanding the project or a decision.
KEY IDEA	Teal. A concept or checkpoint to remember; yellow marks short phrases.
WATCH OUT	Orange. A limitation, assumption, or common mistake to read carefully.
OPTIONAL - FIRST PASS	Purple. Return when you need this detail; the text is still here.

Reading labels are added guidance. The original text, figures, dates, and source references are retained. Page numbers in this edition may differ from the originals.

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How to read this document

This is document C of four. Document A (“Read This First”) explains the system we are building. This one explains the thinking the system automates, in the order Aswath Damodaran teaches it. It is not exam preparation. It is the dense version a friend would give you before you open his lectures again, with a pointer to the packet, session or blog post behind each idea.

Read the seven chapters in file order, one evening each:

1. Why value at all: price versus value, the story-to-numbers frame, the 3P test.
2. The accounting minimum and the base year: where every input number comes from.
3. The DCF chain: what the calculator does, step by step, with the Almarai golden case.
4. Discount rates and country risk: risk-free rate, equity risk premium, betas, cost of debt, India.
5. Growth, terminal value and the dark side: where valuations go wrong and how to show a range.
6. Relative valuation and what the market prices in: multiples, the implied premium, the regime panel.
7. India and the rules of thumb: the currency, tax and data choices, and a one-page checklist.

Every number in the text has a “Source:” line under it that names the file in the local corpus (~\Downloads\financeMD\) or the research report it came from. When a chapter says “the calculator”, it means the fcffsimpleglinzu workbook that the engine reproduces to one part in a million. Nothing here is advice to buy or sell anything; these are estimation methods and their limits.

Where this document stops, document D (“The Other Side”) starts: it collects the people who disagree with him, and what he has said back.

Chapter 1: Why value, the story, and the 3P test

READ FIRST

Start with price versus value, the five-step story, and the 3P test. These explain what the system is trying to estimate and which assumptions belong in its base case.

This is the first chapter of “Damodaran, the important parts”. It is a revision brain-dump from a friend, not a textbook. Every concept ends with one line on why it matters for the calculator we are building. Terms are defined the first time they appear.

Why value anything at all

Valuation is the act of estimating what an asset is worth. An asset is anything that can produce money for its owner in the future, such as a business or a building. Damodaran’s starting rule is that a sensible investor does not pay more for an asset than it is worth. He calls the opposite belief, that any price is fine if someone will pay more later, “a very expensive game of musical chairs”. We own financial assets for the cash they are expected to produce, not for how they make us feel. So the price we pay should be tied to those cash flows, their uncertainty, and their growth.

Source: financeMD/damodaran/New_Home_Page/background/valintro.md (no date in file)

He also warns you about three myths before you touch a spreadsheet. First, valuation is not an objective search for a true value; every valuation is biased, and the only questions are how much and in which direction. Second, no valuation is precise, and the payoff to valuing something is greatest

exactly when the numbers are least precise. Third, a more complicated model is not a better model; your understanding of a model falls as its inputs multiply. The cure is parsimony, which means using the simplest model you can get away with. In his words, “if we can value an asset with three inputs, we should not be using five.”

Source: financeMD/damodaran/pdfiles/eqnotes/ValIntroSpr25.md (Spring 2025);
financeMD/damodaran/New_Home_Page/background/valintro.md (no date in file)

Why it matters for the calculator: the engine must be exact in its arithmetic, but the inputs are guesses, so every run also produces a range from Monte Carlo draws and a sensitivity table. Monte Carlo means running the model many times with inputs drawn at random from a chosen spread. That design is the three myths turned into code.

Source: plans/help-me-out-here-abstract-wilkinson.md (2026-09)

Price versus value

Price and value are two different things that people use as one word. Value is driven by the cash flows an asset produces, the growth in those cash flows, and the risk that they do not show up. Price is driven by demand and supply, which in practice means mood and momentum. Damodaran’s line is that markets are pricing mechanisms, not value mechanisms. Think of a house: the value is what the rent stream is worth, the price is what the neighbor paid last week.

Source: financeMD/damodaran/blog/2020/03/a-viral-market-meltdown-iii-pricing-or.md (2020-03-16)

He draws three consequences. Price has no upper or lower bound, but value has both, set by the most optimistic and most pessimistic sensible forecasts. Price is reactive, moving on each bit of news, while value is proactive, because news must pass through your expectations first. And price may never converge on value in your lifetime, so working on value is an act of faith. His India lecture reduces the subject to three questions: is there a gap between price and value, will it close, and what will close it.

Source: financeMD/damodaran/blog/2020/03/a-viral-market-meltdown-iii-pricing-or.md (2020-03-16);
financeMD/damodaran/pdfiles/country/val2dayIndia2025.md (2025)

Why it matters for the calculator: the calculator computes value, and the market price is a separate input typed in beside it. The output is a gap, nothing more. A rule in the plan scans every memo and answer for advice language and fails the build if it finds any.

Source: plans/help-me-out-here-abstract-wilkinson.md (2026-09)

The three ways to value anything

Damodaran says there are hundreds of valuation models but only three approaches. The table is the whole map.

Approach	What it does	What you need	Works best when	Calculator
Intrinsic valuation (usually DCF)	Value equals the present value of the cash flows the asset will produce	The life of the asset, the cash flows over that life, and a discount rate	You have a long horizon and can wait for price to move	This is the engine itself

Approach	What it does	What you need	Works best when	Calculator
Relative valuation, also called pricing	Estimates what to pay from what others pay for comparable assets, scaled by a shared metric	Comparable assets, a standardized price such as price to earnings, and controls for differences	Many comparables exist and you are judged against a benchmark	A later module; the memo also asks what the current price already requires
Contingent claim, also called real option valuation	Adds value for assets whose payoff depends on an event happening	An underlying asset value, an exercise price, a life, and a measure of variability	Distressed equity, patents, natural resource reserves, rights to expand	Off by default; the “possible” bucket of the 3P test lands here

Some vocabulary. DCF stands for discounted cash flow, which means shrinking future cash by a rate that reflects risk and time. A multiple is a price divided by a common measure, such as earnings, so that companies of different sizes can be compared. An option is a right, not an obligation, to do something later. Two cautions from the same packet stay with me. An intrinsic model can find every stock in a market worth less than its price, which troubles anyone who must stay fully invested. Pricing assumes the market is right on average and wrong only on individual names, so it fails when the whole market is off. And option models sit on top of another valuation, so counting a patent in the growth rate and again as an option is double counting.

Source: [financeMD/damodaran/pdfiles/eqnotes/ValIntroSpr25.md](#) (Spring 2025)

Why it matters for the calculator: the engine is an intrinsic model, so it will disagree with prices for long stretches. That is expected behavior, not a bug.

Story to numbers: the five steps

Damodaran describes two tribes. Number crunchers build spreadsheets and distrust stories; story-tellers pitch visions and distrust spreadsheets. Numbers without a story are just modeling, and a story without numbers is just storytelling. Numbers-only models suffer from three illusions: precision, objectivity, and control. Story-only pitches drift into fantasy and give you no yardstick for progress. A good valuation binds the numbers to **one coherent story**.

Source: [financeMD/damodaran/blog/2014/06/numbers-and-narrative-modeling-story.md](#) (2014-06-24); [financeMD/damodaran/New_Home_Page/NNPreface.md](#) (no date in file)

The process has five steps, and he is not rigid about the order.

1. Develop a narrative for the business. Keep it simple and focused, and make it your own, not management's.
2. Test the narrative against history, experience, and common sense. This is where the 3P test lives; the packet says “No fairy tales or runaway stories.”
3. Convert the key parts of the narrative into drivers of value: market size, market share, margins, reinvestment, and risk.
4. Connect the drivers to a valuation, which for him is a DCF.
5. Keep the feedback loop open. Listen to people who know the business better, and be ready to say “I was wrong”.

Source: [financeMD/damodaran/blog/2014/06/numbers-and-narrative-modeling-story.md](#) (2014-06-24); [financeMD/damodaran/pdfiles/country/val2dayIndia2025.md](#) (2025)

The Uber post shows why the story matters more than the small inputs. In June 2014 he ran four narratives through the same model.

Narrative	Total market	Share	Cost of capital	Value
Car service company facing competitive and regulatory hurdles	100 billion dollars	10 percent	12 percent	3.2 billion dollars
Car service company with sustained profitability (his own)	100 billion dollars	10 percent	12 percent falling to 8	5.9 billion dollars, plus 2 to 3 billion for a disruption option
Car service company with dominant share from network effects	100 billion dollars	50 percent	12 percent falling to 8	29.1 billion dollars
Logistics company expanding into other businesses	600 billion dollars	5 percent	12 percent falling to 8	17.5 billion dollars

His conclusion: “big differences in valuation almost always result from differing narratives”.

Source: [financeMD/damodaran/blog/2014/06/numbers-and-narrative-modeling-story.md](https://financeMD.com/damodaran/blog/2014/06/numbers-and-narrative-modeling-story.md) (2014-06-24)

Why it matters for the calculator: his workbook has a sheet called “Stories to Numbers” with the story in a text box, then one row per assumption with a “Link to story” column. The shipped example carries Amazon’s story text over Almarai’s numbers, and the sheet does not notice, because the story column is free text with no checks. Our memo writer uses the same template but forces every proposed input to cite a filing page or a Damodaran passage, and a second opposing memo per company is mandatory.

Source: [financeMD/damodaran/pc/fcffsimpleginzu.md](https://financeMD.com/damodaran/pc/fcffsimpleginzu.md) (valuation date 2026-02-01 in sheet); [plans/help-me-out-here-abstract-wilkinson.md](https://financeMD.com/plans/help-me-out-here-abstract-wilkinson.md) (2026-09)

The 3P test: possible, plausible, probable

KEY IDEA

This is the filter for the story. Read the definitions and the example before treating a possibility as an input.

The 3P test is the filter for step 2. A possible story could happen. A plausible story could reasonably happen, given how businesses and people behave. A probable story is one you would put weight on today. In his words, “not everything that is possible is plausible, and not all plausible opportunities make the transition to the probable.” Picture three nested circles: the outer one is huge and nearly worthless, the inner one is small and carries the value.

Source: [financeMD/damodaran/blog/2014/07/possible-plausible-and-probable-big.md](https://financeMD.com/damodaran/blog/2014/07/possible-plausible-and-probable-big.md) (2014-07-16)

The test was born in a public disagreement. Bill Gurley, an Uber board member, argued for a far bigger market and strong network effects, which are benefits each user gets from other users joining. Damodaran did not say Gurley was wrong; he said they drew the lines between probable, plausible, and possible in different places. He kept the car ownership market as a possibility with an option value of about 2 to 3 billion dollars, outside the base case. His slides name the two classic failures:

the “Big Market Delusion” for the implausible, and “Willy Wonkitis” for the improbable, his term for pitches built on magical thinking.

Source: [financeMD/damodaran/blog/2014/07/possible-plausible-and-probable-big.md](#) (2014-07-16); [financeMD/valpacket1spr25.md](#) (Spring 2025)

To run the test, use his three checks. History asks whether any company has ever lived this story. Experience asks what road bumps you hit last time you believed a similar story. Common sense asks what economics and arithmetic say about the weakest link. The workbook’s Diagnostics sheet turns this into blunt questions: how big is the total market today, how much do the biggest companies in it make, and what share are you forecasting in year 10.

Source: [financeMD/damodaran/blog/2014/06/numbers-and-narrative-modeling-story.md](#) (2014-06-24); [financeMD/damodaran/pc/fcffsimpleginzu.md](#) (valuation date 2026-02-01 in sheet)

Why it matters for the calculator: the plan hard-codes the test as a rule. Only probable claims may set a base-case input. Plausible claims become a scenario. Possible claims are recorded at zero, which is where an option value would go if we ever add one.

Source: [plans/help-me-out-here-abstract-wilkinson.md](#) (2026-09)

The corporate life cycle lens

The corporate life cycle is the idea that a company ages like a person: young, growth, mature, declining. Value is always cash flows, growth, and risk, but which inputs you sweat over changes with age. He does not invent new models for each stage; “we use the same models” and move the estimation effort.

Source: [financeMD/damodaran/pdfiles/CLC/slides/Ch1.md](#) (no date in file)

Stage	Revenue growth	Margins	Reinvestment	Cost of capital and failure	Where the value sits
Young or start-up	The one number you can see	Negative, slow to turn positive	From sales-to-capital, the revenue each unit of capital supports	Split operating risk from failure risk; failure is its own probability	Almost all in the future
High growth	High, fading toward the sector	Rising as growth spending stops hiding profit	Sales-to-capital again, with lags allowed	A year-specific cost of capital that falls as growth moderates	Terminal value can be 80 or 90 percent of value, or more than 100 percent
Mature	Close to the economy's growth rate	Stable; use historic norms	Shifts toward acquisitions; ask whether they create value	An established financing mix; failure risk near zero	In assets already in place; financing and dividend policy matter more
Declining	Flat or negative, often from selling assets	Compressing	Can be negative, because divestitures bring cash in	Distress raises the cost of debt and equity; failure probability rises	Depends on the management path: denial, desperation, acceptance, or reinvention

Source: financeMD/damodaran/pdfiles/CLC/slides/Ch10.md, Ch11.md, Ch12.md, Ch13.md (no date in files)

A few details are worth keeping. Venture capitalists price young companies with target rates of return, 50 to 70 percent for a start-up and 25 to 35 percent at the bridge or IPO stage. Damodaran calls that a forward pricing, not a valuation, because the target rate mixes operating risk and failure risk with no stated logic. His Zomato example instead uses a rupee cost of capital of 10.25 percent falling to about 9 percent, plus a 10 percent chance of failure. His Unilever example uses a cost of capital of 8.97 percent, 78 percent equity and 22 percent debt, and no failure risk. Terminal value, the value of all cash flows after the forecast period, is the input to watch in growth companies. The 3P test is mostly unnecessary for a mature company whose story extends its history, with two exceptions: imminent macro or regulatory change, and disruption.

Source: financeMD/damodaran/pdfiles/CLC/slides/Ch10.md, Ch11.md, Ch12.md (no date in files)

The preface to Narrative and Numbers gives the manager's side. Early on, the masterful storytellers win; as the company grows, the skill shifts to delivering numbers that back the story; a mature company's narrative is "mostly written"; a declining company's is about finding a decent ending.

Source: financeMD/damodaran/New_Home_Page/NNPreface.md (no date in file)

Why it matters for the calculator: the workbook's five levers (growth next year, growth for years 2 to 5, target margin, years to converge, sales-to-capital) plus its failure-probability switch are how the life cycle enters the numbers. The memo must also name the company's uncertainty type so the right chapter of "The Dark Side of Valuation" is pulled before inputs are proposed.

Source: claude/tool-results/bkdrkpfng.txt (2026-09); plans/help-me-out-here-abstract-wilkinson.md (2026-09)

Narrative breaks, shifts, and tweaks

A story is never finished, because the world delivers surprises. Damodaran sorts them into three kinds and pairs each with a tool.

Kind	What happened	Effect on your valuation	Tool
Narrative break or end	A legal, political, economic, or credit event makes the story no longer operative	The old cash flows, growth, and risk numbers are void	Estimate a probability of the break and its consequences
Narrative shift	The business model works better or worse than expected; market size, share, or margins move	Update the inputs with the new data	Monte Carlo simulation or scenario analysis
Narrative change, expansion or contraction	Unexpected entry into a new market, or exit from an existing one	Redo the valuation with a new market potential	Real options

Source: financeMD/damodaran/blog/2014/08/reacting-to-earnings-reports-narrative.md (August 2014); financeMD/valpacket1spr25.md (Spring 2025)

The book chapter is titled "Narrative breaks, shifts and tweaks", and a tweak is the smallest move: keep the story, nudge a number. The label depends on your starting story. Uber succeeding in the suburbs was a change for Damodaran, whose base case was urban, but only a shift for Gurley, whose base case already included suburbs. His warning is that a rigid DCF responds to all three with

denial, keeping inputs fixed and blaming the market. The right posture is proactive: bring failure in as a probability for young or indebted firms, use distributions where shifts are likely, and treat true changes as options. And seek out “people who think least like you”.

Source: financeMD/damodaran/New_Home_Page/NNPreface.md (no date in file);
 financeMD/damodaran/blog/2014/08/reacting-to-earnings-reports-narrative.md (August 2014);
 financeMD/damodaran/pdfiles/CLC/slides/Ch10.md (no date in file)

Why it matters for the calculator: every run is stored with its inputs and data vintage, so a shift is a new run with changed levers, a break is the failure-probability switch, and a change is a new memo with a new story. The nightly jobs re-value the watchlist, so the feedback loop is a schedule rather than a good intention.

Source: plans/help-me-out-here-abstract-wilkinson.md (2026-09)

Where this comes from

Go deeper in this order. Paths are under financeMD/ unless shown in full.

- damodaran/New_Home_Page/background/valintro.md: the standalone primer on bias, uncertainty, complexity, and the three approaches.
- damodaran/pdfiles/eqnotes/ValIntroSpr25.md: the Spring 2025 class opener; myths and truths, the three approaches side by side.
- valpacket1spr25.md (corpus root), slides 264 to 272: the narrative session, the Big Market Delusion, Willy Wonkitis, and the break, shift, and change table.
- damodaran/pdfiles/country/val2dayIndia2025.md, slides 4 to 9 and 148 to 149: the price-versus-value gap and the five steps, taught in India.
- damodaran/blog/2014/06/numbers-and-narrative-modeling-story.md: the original five steps and the four Uber narratives.
- damodaran/blog/2014/07/possible-plausible-and-probable-big.md: the 3P test, argued against Bill Gurley.
- damodaran/blog/2014/08/reacting-to-earnings-reports-narrative.md: breaks, shifts, and changes, with the tool for each.
- damodaran/blog/2020/03/a-viral-market-meltdown-iii-pricing-or.md: price versus value, written during a crash.
- damodaran/pdfiles/CLC/slides/Ch1.md and Ch10.md to Ch13.md: the life-cycle lens with the Zomato, Unilever, and declining-firm examples.
- damodaran/pc/fcfsimpleginzu.md, the “Stories to Numbers” and “Diagnostics” sheets: the template the memo writer imitates.

Three things to remember

KEY IDEA

1. Value comes from cash flows, growth, and risk; price comes from demand and supply. The calculator computes the first, reports the gap to the second, and never turns that gap into advice.
2. Every number must be tied to a sentence in a story, and every sentence must pass the 3P test. **Only the probable sets the base case**; the plausible gets a scenario; the possible is written down at zero.
3. The model does not change with a company's age, but the inputs you fight over do. Young companies live or die on growth and failure risk, mature ones on financing and dividend policy, declining ones on the path management takes.

Chapter 2: The Accounting Minimum and the Base Year

READ FIRST

Focus on where the starting numbers come from. Learn the three statements, the expense categories, and the updating rule before worrying about every accounting adjustment.

What this chapter is

Chapter 3 walks the calculator forward from a base year. This chapter is about that base year: the handful of numbers you type in before anything is forecast. Damodaran's view of accounting is simple and slightly rude. Accountants record the past carefully, and valuers want the future, so the statements are a starting point that needs repairs. You do not need to become an accountant. You need to know three statements, one distinction, two repairs, two tax rates, and one updating rule.

Source: [financeMD/damodaran/New_Home_Page/AccPrimer/accstate.md](https://financeMD.com/damodaran/New_Home_Page/AccPrimer/accstate.md) (undated primer)

The three statements and how they connect

A company publishes three tables. The balance sheet is a photograph: what the company owns (assets) and who paid for it (liabilities and equity) on one date. The income statement is a film: revenues earned and expenses used up over a period, ending in net income. The cash flow statement is the bank record: cash that moved in and out during the same period, split into operating, investing and financing activities.

They connect in two places. Net income from the income statement flows into equity on the balance sheet as retained earnings. The cash flow statement starts from net income, adds back non-cash charges such as depreciation, and ends at the change in the balance sheet's cash line. If you remember one link, remember this one: profit is an opinion, cash is a fact, and the cash flow statement is the reconciliation between them.

Why it matters for the calculator: every base-year input comes from one of these three tables, and the table tells you whether the number is a stock (a balance on a date) or a flow (a total over a period).

Source: [financeMD/Aswath Damodaran - Investment Valuation, University Edition _ Tools and Techniques for Determining the Value of Any Asset \(2023\).md](https://financeMD.com/Aswath_Damodaran_-_Investment_Valuation,_University_Edition_-_Tools_and_Techniques_for_Determining_the_Value_of_Any_Asset_(2023).md), Chapter 3, "Understanding Financial Statements", Figures 3.1 to 3.3

Operating, financing, capital: the one distinction that matters

Damodaran sorts every expense into three bins. An operating expense buys benefits only in the current period; wages and raw materials are the classic cases. A financing expense is the cost of borrowed money, chiefly interest. A capital expense buys benefits over many years, such as a factory, and is spread across those years through depreciation.

The income statement is built in that order. Revenues minus operating expenses minus depreciation gives operating income, also called EBIT (earnings before interest and taxes). EBIT minus interest gives taxable income. Taxable income minus taxes gives net income. The trouble is that accountants put some items in the wrong bin. Research spending is a capital expense wearing an operating-expense costume. Lease rent, before 2019, was a financing expense in the same costume. He describes a financial expense as “any commitment that is tax deductible that you have to meet no matter what your operating results”.

Why it matters for the calculator: the calculator forecasts operating income and treats interest separately, so anything mis-binned in the base year is mis-forecast for ten years.

Source: [financeMD/damodaran/blog/2011/06/from-revenues-to-earnings-operating.md](#) (2011-06-15); [financeMD/valpacket1spr25.md](#), slides 121 to 122

Cash versus accrual

Accrual accounting records a sale when the good is delivered, not when the cash arrives. It then matches the expenses that produced that sale to the same period. A cash basis would record only money that moved. Accrual is the better measure of performance, which is why the income statement uses it. But it also means revenue can be booked before it is collected, and inventory or receivables can hide the gap.

The practical consequence is that free cash flow, which is what the calculator discounts, must be rebuilt from accrual earnings. You start with after-tax operating income and subtract reinvestment. That is the bridge from accrual to cash, and chapter 3 covers the mechanics.

Why it matters for the calculator: the base-year EBIT is an accrual number, and the calculator turns it into cash by subtracting reinvestment, so the base EBIT must be clean of one-off items.

Source: [financeMD/damodaran/New_Home_Page/AccPrimer/accstate.md](#), “How Accountants measure earnings”; [financeMD/damodaran/New_Home_Page/AccPrimer/onetime.md](#)

Two repairs: capitalize R&D, and operating leases before 2019

Research and development

Accounting rules force companies to expense R&D in the year it is spent. Damodaran argues it is a capital expense and should be capitalized, meaning turned into an asset and depreciated. The recipe has three steps. Choose an amortizable life, two to ten years depending on the industry. Collect past R&D for that many years. Sum the unamortized portions to get a research asset, and compute this year's amortization of it.

The adjustment to operating income is this year's R&D minus this year's amortization. In his worked example, in euros and with a five-year life, R&D of 1,020 and amortization of 903 raised operating income by 117 and created a research asset of 2,914. That asset is added to book equity and to invested capital, so return on capital changes as well.

Why it matters for the calculator: the workbook's "R&D converter" sheet does exactly this, and the engine must add the adjustment to base-year EBIT and the research asset to invested capital when the R&D switch is "Yes".

Source: financeMD/valpacket1spr25.md, slides 130 to 132;
financeMD/damodaran/New_Home_Page/AccPrimer/research.md; bkdrkpfnfng.txt, section 3.7

Operating leases

Before 2019 an operating lease, meaning a rental contract for a store or an aircraft, stayed off the balance sheet. The rent sat inside operating expenses. Damodaran treated the rent as a financing expense, discounted the future lease commitments at the pre-tax cost of debt, and called that present value debt. Operating income was then raised by the rent and lowered by depreciation on the created lease asset. In his Gap example the debt value of leases was 4,397 against conventional debt of 1,970, and adjusted operating income rose from 1,012 to 1,362.

In 2019 both IFRS and US GAAP changed the rules, so companies now report lease liabilities and right-of-use assets themselves. The workbook carries a warning: do not run the lease converter on a post-2019 filing, because the accountant has already done it and you would double count.

Why it matters for the calculator: for modern filings the lease switch stays "No", and the decision moves to whether reported lease liabilities are counted inside book debt.

Source: financeMD/valpacket1spr25.md, slides 123 to 126;
financeMD/damodaran/New_Home_Page/AccPrimer/lease.md; bkdrkpfnfng.txt, section 1.2, row B17

Effective versus marginal tax rate, and NOLs

The effective tax rate is taxes divided by taxable income, as reported in the statements. It is an average across everything the company earned, including low-tax foreign profits and deferrals. The **marginal tax rate** is the statutory rate on the next unit of profit, found in the tax code of the home country. Most companies report an effective rate below their marginal rate.

Damodaran's rule is to start with the effective rate and drift toward the marginal rate over the forecast. Using only the marginal rate understates early cash flows. Using only the effective rate assumes the tax deferral lasts forever. The marginal rate is the right one for the tax benefit of interest, because interest is deducted from the last dollars of income. In his January 2026 country data, India shows an average effective rate of 22.33% against a marginal rate of 30%.

An NOL is a net operating loss carried forward: a bank of past losses that shields future profits from tax until it is used up. The workbook has an override to enter an opening NOL, and the tax computation skips tax on income until the bank is exhausted. The engine will port that logic, as chapter 3 shows.

For India the marginal-rate options are 25.17% under the new regime, 34.94% under the old regime, and 17.16% for new manufacturing companies. The workbook's country table lists 30% for India, but no formula reads it, so both tax rates are typed by hand.

Why it matters for the calculator: two tax inputs, effective for years 1 to 5 and marginal for the terminal year and the cost of debt, and an optional NOL bank.

Source: financeMD/valpacket1spr25.md, slides 138 to 140;
financeMD/damodaran/New_Home_Page/definitions.md, rows "Effective tax rate" and "Marginal tax rate";
financeMD/damodaran/blog/2026/02/data-update-7-for-2026-debt-and-taxes.md (2026-02-20);
bkdrkpfnfng.txt, sections 1.2 and 6

The base-year inputs, and where each one lives

WATCH OUT

Check the units as well as the source. The table links each starting input to the financial statements; the share count must use compatible units.

The workbook's Input sheet asks for these numbers, in two columns: the most recent twelve months, and the last annual report before that. **Units must be consistent** all the way through, including the share count. Apple's 2025 10-K reports in millions of dollars; Trent's Indian results report in rupees crore, where one crore is ten million.

The table below shows where each item sits. The Apple figures come from the fiscal year ended September 27, 2025. The Trent figures come from the standalone results for the year ended March 31, 2024, which is the annual column inside a quarterly filing.

Input	What it is	US 10-K: Item 8, Apple 2025	Indian filing: Trent, Rs crore
Revenues	Sales earned in the period	Statement of Operations, "Total net sales" 416,161	"Revenue from operations" 11,926.56; exclude "Other income"
Operating income (EBIT)	Revenues minus operating costs and depreciation	"Operating income" 133,050	Not printed; compute profit before tax 1,873.32 plus finance costs 309.37 minus other income and exceptional items
Interest expense	Cost of borrowed money	Not broken out; sits inside "Other income/(expense), net" of (321); estimate from the debt note	"Finance costs" 309.37
Book value of equity	Paid-in capital plus retained earnings	Balance Sheet, "Total shareholders' equity" 73,733	"Paid-up equity share capital" 35.55 plus "Other equity" 4,411.64, printed as "Net Worth" 4,447.19
Book value of debt	Interest-bearing borrowings	Commercial paper 7,979 plus term debt 12,350 and 78,328; lease liabilities 13,720 are in the lease note	"Paid up Debt capital" 1,738.32, which by its own note includes lease liabilities under Ind AS 116
Cash and marketable securities	Cash and near-cash investments	Cash 35,934 plus marketable securities 18,763 and 77,723	Balance sheet, cash and bank balances plus investments; not in the quarterly results
Cross holdings	Stakes in other companies, not consolidated	"Other non-current assets" and the investments note	Associates and joint ventures, listed in the auditor's report
Minority interests	Outsiders' share of consolidated subsidiaries	Balance Sheet, "Noncontrolling interests" when present; Apple has none	Consolidated results, "Non controlling interest"
Shares outstanding	Count of shares	Cover page: 14,776,353,000 as of October 17, 2025	Share capital 35.55 crore at face value Re 1, so 35.55 crore shares

Input	What it is	US 10-K: Item 8, Apple 2025	Indian filing: Trent, Rs crore
Price	Market price per share	Not in the filing; take it from the market on the valuation date	Same
Effective tax rate	Taxes divided by pre-tax income	Management's Discussion and Analysis (MD&A, Item 7) tax table: 15.6%; statutory federal 21%	"Total tax expenses" 437.50 divided by profit before tax 1,873.32

Four things to notice. First, Apple's 2025 10-K does not print interest expense anywhere obvious, which is why the calculator's interest input sometimes has to be estimated from the debt note. Second, the operating income line is printed in a US 10-K but not in an Indian results statement, so the engine must rebuild it. Third, Indian "Other income" is interest and dividend income, not operations, and must be kept out of both revenue and EBIT. Fourth, Damodaran's definition of debt includes lease commitments, so the Apple lease liabilities of 13,720 belong in book debt if you follow him, and Trent's filing already does that.

Why it matters for the calculator: the memo pipeline will cite a filing page for every one of these inputs, and the nightly job will fill the same table from EDGAR (the SEC's free filing database) for US names and later from EODHD (a paid market-data feed) for Indian names.

Source: financeMD/Apple-2025-10-K.md, Item 8 pages 29 to 33 and the leases note; financeMD/Unaudited Financial Results (Standalone and Consolidated) - Q1 FY 2024.md (filed 2024-08-09); financeMD/damodaran/New_Home_Page/definitions.md, row "Debt"; bkdrkpfng.txt, section 1.2

On cross holdings and minority interests

A minority interest, now called a non-controlling interest, appears when a company owns more than half of a subsidiary and consolidates all of it. The outsiders' slice is shown as a liability and must be subtracted from firm value. A cross holding below fifty percent is not consolidated; its book value sits among the assets and must be added back. Both are book values, and Damodaran converts them to market value with a sector price-to-book ratio when he can. Emerging market companies carry more cross holdings, often to preserve family control.

Why it matters for the calculator: two inputs, minority interests subtracted and cross holdings added, both in the equity bridge of chapter 3 (the steps that turn firm value into value per share).

Source: financeMD/valpacket1spr25.md, slides 227 to 229 and the emerging-market cross-holdings slides; financeMD/damodaran/New_Home_Page/definitions.md, row "Minority Interests"

The trailing-twelve-month rule

Annual reports go stale. A company valued in April is working from numbers that ended the previous September. The fix is **trailing-twelve-month data**, meaning the most recent twelve months built from one annual report and one interim report. The rule is: last annual figure, minus the same interim period of the prior year, plus the current interim period.

Damodaran's own Apple example makes it concrete. Annual revenue to September 2023 was 383,285. The six months to March 2023 were 211,990 and the six months to March 2024 were 210,328. Trailing revenue is 383,285 minus 211,990 plus 210,328, which is 381,623. Operating income by the same rule went from 114,301 to 118,240. The workbook ships a small "Trailing 12 month Worksheet" that

does this for revenues, operating income, interest expense and the effective tax rate, but it is not wired to the Input sheet; you paste the results by hand.

The rule applies only to flows. Balance sheet items such as cash, debt and equity are stocks, so you simply take the latest interim balance. Some items, such as employee options, appear only in annual reports, so you carry them forward and accept a small inconsistency.

Why it matters for the calculator: the nightly job will compute exactly this, last annual minus prior interim plus current interim, for the flow inputs, and will store both columns with their vintage.

Source: financeMD/Aswath Damodaran - Investment Valuation, University Edition _ Tools and Techniques for Determining the Value of Any Asset (2023).md, Chapter 9, Illustration 9.1; financeMD/valpacket1spr25.md, slide 120; financeMD/damodaran/pc/fcfsimpleginzu.md, “Trailing 12 month Worksheet”; bkdrkpfng.txt, section 3.8

Where this comes from

- financeMD/damodaran/New_Home_Page/AccPrimer/accstate.md, the primer on financial statements, the shortest full statement of his view of accounting.
- financeMD/damodaran/New_Home_Page/AccPrimer/research.md, lease.md and onetime.md, the three repairs in his own words.
- financeMD/damodaran/New_Home_Page/definitions.md, one-line definitions of every term in this chapter, with his remarks.
- financeMD/valpacket1spr25.md, slides 119 to 141, the “accounting earnings, flawed but important” run and the tax slides.
- financeMD/cfpacket1spr25.md, the financial balance sheet framing: assets in place, growth assets, debt and equity.
- financeMD/damodaran/blog/2011/06/from-revenues-to-earnings-operating.md, the Groupon post on operating, financing and capital expenses.
- financeMD/damodaran/blog/2010/03/accounting-inconsistencies.md, the opening of his series on accounting inconsistencies.
- financeMD/damodaran/blog/2026/02/data-update-7-for-2026-debt-and-taxes.md, marginal, effective and cash tax rates by country.
- financeMD/Aswath Damodaran - Investment Valuation, University Edition _ Tools and Techniques for Determining the Value of Any Asset (2023).md, Chapter 3 and Chapter 9.
- financeMD/Apple-2025-10-K.md and the Trent results file, as worked examples of where numbers live.

Three things to remember

KEY IDEA

1. Sort every expense into operating, financing or capital, then fix the two that accountants mis-bin: capitalize R&D, and treat lease commitments as debt unless the filing already does.
2. Start at the effective tax rate and fade to the marginal rate; use the marginal rate for the tax benefit of debt; give loss-makers an NOL bank.
3. Flows are trailing twelve months, last annual minus prior interim plus current interim; stocks are the latest balance; units stay consistent through the share count.

Chapter 3: The DCF Chain, Exactly As the Workbook Does It

READ FIRST

Follow Almarai through all nine stations. First trace the sequence; then return to the formulas. The closing three points give you a compact way to check your understanding.

What this chapter is

A DCF (discounted cash flow) valuation is a way of putting a number on a business today. You forecast the cash the business will throw off each year. Then you shrink each year's cash to allow for waiting and for risk. The `fcffsimpleginzu` workbook does this in one fixed chain of steps. This chapter walks that chain in the order the spreadsheet computes it. Think of a conveyor belt with **nine stations**. Revenue goes in at one end. A value per share comes out at the other.

We follow one company through every station: Almarai, a Saudi food company, valued on 2026-02-01. It is the example that ships inside the workbook. Its answer, 7.187840270062114 per share against a share price of 72.28, is the golden test for your calculator. The calculator must reproduce that number to six decimals before it is trusted with anything else. All Almarai figures below are in millions of Saudi riyals, except the share count and the per-share numbers.

Source: `plans/help-me-out-here-abstract-wilkinson.md` (2026-09);
`financeMD/damodaran/pc/fcffsimpleginzu.md` (valuation date 2026-02-01)

The grid you are filling in

The “Valuation output” sheet is a grid. One column is the base year, meaning the last twelve months of actual results. Ten columns are forecast years 1 to 10. One more column is the terminal year, the first year of “forever”. Every station below fills one or two rows of that grid.

Almarai's starting inputs, as typed on the Input sheet:

Input	Almarai value
Revenues, last twelve months	21,765.4
Operating income (EBIT)	3,060.9
Effective tax rate, then marginal tax rate	17.5%, then 25%
Book value of debt	45,063
Cash and marketable securities	19,000
Cross holdings and other non-operating assets	21,119
Minority interests	1,558
Shares outstanding	4,315
Share price	72.28
Risk-free rate	4.58%
Mature-market equity risk premium	4.23%
Cost of capital, years 1 to 5	7.055%
Sales to capital ratio	1.7085

EBIT means earnings before interest and taxes. It is the profit from running the business, before lenders and the tax office are paid. Every other term in the table gets defined at the station where it is used.

Source: claudetool-results/bkdrkpfng.txt, section 1 (2026-09)

Station 1: the revenue growth path

Revenue is the money customers pay the company. The workbook needs only three growth inputs. Year 1 growth is cell B26. Growth for years 2 to 5 is cell B28, and it is one flat rate. Terminal growth is the growth rate in “forever”, and by default it equals the risk-free rate (station 7 explains why). Years 6 to 10 are not typed at all. They glide from the year-5 rate to the terminal rate in five equal steps, so year 10 lands exactly on the terminal rate. Picture a plane descending on a straight glide path to the runway.

Almarai uses 5% for year 1 and 5% for years 2 to 5. The terminal rate is 4.58%. The step is (5% minus 4.58%) divided by 5, which is 0.084 percentage points a year. So years 6 to 10 grow at 4.916%, 4.832%, 4.748%, 4.664% and 4.580%. Revenue climbs from 21,765.4 to 22,853.7 in year 1, 27,778.8 in year 5, 35,030.0 in year 10 and 36,634.4 in the terminal year.

```
g_1      = B26
g_2 .. g_5 = B28
g_(5+k)   = g_5 - (g_5 - g_T) / 5 * k      for k = 1..5
Rev_t     = Rev_(t-1) * (1 + g_t)
Rev_T     = Rev_10 * (1 + g_T)
```

Why it matters for the calculator: compute all eleven revenue columns first, because station 4 looks one year ahead.

Source: bkdrkpfng.txt, section 2.2; financeMD/damodaran/pc/fcfsimpleginzu.md, “Valuation output” rows “Revenue growth rate” and “Revenues” (2026-02-01)

Station 2: operating margin convergence

Operating margin is EBIT divided by revenue, the share of each sale left after operating costs. Three inputs control it. Year 1 margin (B27) defaults to the base-year margin. The target margin (B29) defaults to the year 1 margin. The year of convergence (B30) defaults to 5. Between year 1 and the convergence year, the margin moves in equal steps toward the target. After that year it stays at the target. Think of a thermostat set to a temperature and a fixed number of minutes to get there.

```
m_1 = B27
m_t = target - ((target - m_1) / Y) * (Y - t)   for t <= Y   (Y = B30, target = B29)
m_t = target                                   for t > Y
EBIT_t = m_t * Rev_t
```

Almarai’s base margin is 3,060.9 divided by 21,765.4, which is 14.063%. The defaults were left untouched, so the margin line is flat at 14.063% in every year. EBIT is therefore just 14.063% of each year’s revenue: 3,213.9 in year 1 and 4,926.3 in year 10.

Why it matters for the calculator: B27 and B29 are pre-filled formulas that the user is meant to overwrite. Store them as initial values, not permanent links.

Source: bkdrkpfng.txt, sections 1.3 and 2.3; fcfsimpleginzu.md, “Valuation output” rows “EBIT (Operating margin)” and “EBIT (Operating income)” (2026-02-01)

Station 3: the tax fade

The effective tax rate is what the company actually paid, as a share of pre-tax profit. The marginal tax rate is the statutory rate on the next unit of profit. The workbook applies the effective rate for years 1 to 5. In years 6 to 10 the rate fades in five equal steps to the marginal rate. The terminal year uses the marginal rate. The logic is that tax breaks run out as a company matures.

Almarai pays 17.5% in years 1 to 5. The step is (25% minus 17.5%) divided by 5, which is 1.5 points a year. So years 6 to 10 are 19%, 20.5%, 22%, 23.5% and 25%. After-tax operating income, written EBIT(1-t), is 2,651.5 in year 1 and 3,694.7 in year 10.

There is also an NOL branch. NOL means net operating loss carryforward, a bank of past losses that shields future profit from tax. Losses are not taxed. If the NOL bank is bigger than this year's EBIT, no tax is paid. Otherwise only the excess is taxed. Almarai's NOL is zero in every year. The terminal year ignores the NOL bank.

Why it matters for the calculator: port both the linear fade and the NOL shield, because the model will meet loss-making companies.

Source: bkdrkpfng.txt, section 2.4; fcffsimpleginzu.md, "Valuation output" rows "Tax rate", "EBIT(1-t)" and "NOL" (2026-02-01)

Station 4: reinvestment, or why growth is not free

KEY IDEA

Growth needs funding. Follow how extra revenue becomes reinvestment before it becomes free cash flow.

Reinvestment is the money a company plows back to grow: new plants, more inventory, more credit extended to customers. Invested capital is the money already tied up in the business, measured as book equity plus book debt minus cash. The sales to capital ratio is revenue produced per unit of invested capital. A ratio of 2 means one rial of capital supports two riyals of sales.

The workbook does not forecast capital spending line by line. It asks how much extra revenue you want next year, then divides by the sales to capital ratio. That is the capital you must put in this year. The one-year lag is the point: you build the shop before you sell from it.

$$\begin{aligned} \text{Reinv}_t &= (\text{Rev}_t(t+1) - \text{Rev}_t) / \text{s2c}_t && (\text{default lag} = 1) \\ \text{InvCap}_t &= \text{InvCap}_t(t-1) + \text{Reinv}_t \\ \text{ROIC}_t &= \text{EBIT}(1-t)_t / \text{InvCap}_t(t-1) \end{aligned}$$

Almarai year 1: (23,996.4 minus 22,853.7) divided by 1.7085 gives 668.8. By year 10 the figure is 939.0. Invested capital rises from 36,730.8 to 44,796.6. ROIC (return on invested capital, after-tax operating income divided by the prior year's capital) rises from 6.875% in the base year to 8.42% in year 10.

Now the catch. The default sales to capital ratio, 1.7085, is the Global industry average for Food Processing, pulled from a lookup table. Almarai's own ratio is 21,765.4 divided by 36,730.8, which is 0.59. The Diagnostics sheet prints both side by side. If you believed the company's own history, each unit of growth would cost nearly three times the capital the default assumes. Damodaran's own words on this: "Growth is not free and it has to be paid for with reinvestment".

Why it matters for the calculator: the lag switch accepts 0 to 3 years, there is no floor, and shrinking revenue produces negative reinvestment. Record that the default came from the Global table.

Source: bkdrkpfng.txt, sections 1.3, 2.5 and 4; fcffsimpleginzu.md, “Valuation output” rows “Reinvestment”, “Invested capital”, “ROIC” and “Diagnostics” step 4 (2026-02-01);
financeMD/damodaran/blog/2016/11/myth-53-growth-is-good-more-growth-is.md (2016-11-30)

Station 5: free cash flow to the firm

FCFF stands for free cash flow to the firm. It is after-tax operating income minus reinvestment. “To the firm” means before any debt payments, so it belongs to lenders and shareholders together. Damodaran’s definitions page writes it as $EBIT(1-t)$ minus net capital spending minus the change in working capital. The workbook collapses those last two items into the single reinvestment line from station 4.

Almarai’s FCFF is 1,982.7 in year 1 and 2,755.7 in year 10. It grows more slowly than EBIT because the tax fade takes a bigger bite each year.

Why it matters for the calculator: FCFF is one subtraction per column. All the difficulty lives upstream.

Source: financeMD/damodaran/New_Home_Page/definitions.md, row “Free Cash Flow to Firm (FCFF)”; fcffsimpleginzu.md, “Valuation output” row “FCFF” (2026-02-01)

Station 6: the cost of capital path and cumulative discount factors

The cost of capital, often called WACC, is the blended annual return that lenders and shareholders together require. Chapter 4 builds Almarai’s starting figure of 7.055%. The workbook holds it flat for years 1 to 5. The terminal cost of capital is the risk-free rate plus the mature-market equity risk premium: 4.58% plus 4.23%, which is 8.81%. Years 6 to 10 glide there in five equal steps of 0.351 points: 7.406%, 7.757%, 8.108%, 8.459% and 8.81%. The idea is that a maturing company drifts toward the risk of an average company.

Because the rate changes each year, you cannot discount with one rate raised to a power. Instead the sheet keeps a cumulative discount factor. Each year’s factor is last year’s factor divided by $(1 + \text{this year's rate})$. Almarai’s factor is 1 divided by 1.07055, or 0.9341, in year 1, and 0.4816 in year 10. Present value of each year’s FCFF is FCFF times that factor: 1,852.0 in year 1, 1,327.2 in year 10. The ten present values add up to 16,394.5.

$$\begin{aligned} \text{CumDF}_1 &= 1 / (1 + \text{WACC}_1) \\ \text{CumDF}_t &= \text{CumDF}_{(t-1)} / (1 + \text{WACC}_t) \\ \text{PV}_t &= \text{FCFF}_t * \text{CumDF}_t \end{aligned}$$

Why it matters for the calculator: the Input sheet label says the terminal rate is “riskfree + 4.5%”. The formula actually adds the mature-market premium from the Country ERP sheet. Code the formula, not the label.

Source: bkdrkpfng.txt, sections 1.6 and 2.6; fcffsimpleginzu.md, “Valuation output” rows “Cost of capital”, “Cumulated discount factor” and “PV(FCFF)” (2026-02-01)

Station 7: terminal value

WATCH OUT

Slow down at the terminal year. Read the three defaults and how terminal reinvestment is recomputed; the final column carries long-lived assumptions.

Terminal value is one number that stands for every cash flow after year 10. The workbook uses the perpetuity formula: terminal cash flow divided by (terminal cost of capital minus terminal growth). Three defaults feed it.

First, terminal growth equals the risk-free rate. The packet's rule: "The stable growth rate cannot exceed the growth rate of the economy, but it can be lower." The risk-free rate is expected inflation plus an expected real interest rate. Nominal economic growth is expected inflation plus expected real growth. So the risk-free rate is a handy ceiling, in the same currency as the cash flows.

Second, the stable return on capital equals the terminal cost of capital. In words, the company stops earning more than its capital costs. The packet: "The excess returns at stable growth firms should approach (or become) zero." Terminal reinvestment then follows the packet's stable-period rule, reinvestment rate = g divided by ROC, and the sheet multiplies that rate by terminal after-tax income.

Third, the terminal cost of capital is risk-free plus mature-market premium, as in station 6.

Almarai's terminal year: revenue 36,634.4, EBIT 5,151.9, after-tax income at 25% of 3,864.0. The reinvestment rate is 4.58% divided by 8.81%, or 51.99%, so reinvestment is 2,008.7. Note the jump from 939.0 in year 10 to 2,008.7. That is not a bug. The formula switches from sales to capital to g divided by ROC. FCFF is 1,855.2. Terminal value is 1,855.2 divided by (8.81% minus 4.58%), which is 43,858.8. Multiplied by the year-10 factor of 0.4816, its present value is 21,123.0.

Adding the two pieces: 16,394.5 plus 21,123.0 gives 37,517.5. By my arithmetic the terminal piece is about 56% of the total. Damodaran's terminal value post argues this is normal, because most equity returns come from price appreciation rather than dividends. His warning cuts the other way too: high terminal share does not mean growth-phase assumptions matter less, because they set the level the terminal value grows from.

Why it matters for the calculator: the "Stories to Numbers" sheet labels the after-year-10 cell "Sales to Capital" but it holds 0.5199, the reinvestment rate. Name it correctly in code.

Source: financeMD/valpacket1spr25.md, slides 201 to 211 (Spring 2025); bkdrkpfng.txt, sections 2.5 and 2.7; fcffsimpleginzu.md, "Valuation output" and "Stories to Numbers" (2026-02-01); financeMD/damodaran/blog/2016/11/myth-55-terminal-value-ate-my-dcf.md (2016-11-30)

Station 8: failure probability and truncation

A DCF assumes the company survives to enjoy its terminal value. The packet says so directly: a DCF "values a firm as a going concern". If failure is plausible, the sheet blends two outcomes. Value of operating assets equals the DCF sum times (1 minus the probability of failure), plus the distress proceeds times that probability. Proceeds are a percentage (B54) of either book capital, code "B", or the DCF value itself, code "V" (B53).

Almarai's probability is 0, so the DCF sum of 37,517.5 passes through unchanged. The proceeds cell still computes 18,758.8, which is 50% of 37,517.5, and then multiplies it by zero. The Failure Rate worksheet holds rating-based default tables and survival-by-age tables, but has no formula links. You read a number off it and type it into B52.

Why it matters for the calculator: truncation is one multiplication after the sum, but B53 and B54 are always read, even when the switch says No.

Source: valpacket1spr25.md, slide 310 (Spring 2025); bkdrkpfng.txt, sections 1.6, 2.7 and 3.5; fcffsimpleginzu.md, "Valuation output" rows "Probability of failure" and "Proceeds if firm fails" (2026-02-01)

Station 9: the equity bridge to value per share

KEY IDEA

Finish the bridge. Operating value becomes equity value only after the other claims and non-operating assets are accounted for. Follow every row of this worked example.

The DCF sum values the operating business. Shareholders own what is left after other claims, plus assets the operations did not use. The bridge for Almarai:

Step	Almarai
Value of operating assets	37,517.5
Minus debt	45,063
Minus minority interests	1,558
Plus cash	19,000
Plus non-operating assets	21,119
Equals value of equity	31,015.5
Minus value of employee options	0
Divided by shares	4,315
Value per share	7.1878
Share price	72.28

Minority interests are the slice of consolidated subsidiaries owned by outsiders, so it is subtracted. Non-operating assets are things like stakes in other companies, which produce no revenue in the forecast, so they are added at book. Employee options are rights to buy new shares cheaply, and they dilute existing holders. The sheet values them with a dilution-adjusted Black-Scholes model that is circular, which is why the workbook needs Excel's iterative calculation switched on. With the options switch set to No, that line is zero. Trapped cash, override number 9, can reduce the cash line by the tax due on bringing it home.

Two honest observations. Almarai's debt, 45,063, is larger than its operating asset value, 37,517.5. The equity value is carried by cash and cross holdings. The Diagnostics sheet computes value divided by price as 0.0994 and prints "Value seems low. See below". Treat Almarai as the arithmetic fixture, not a study of the company. Also, the "Stories to Numbers" sheet still carries leftover Amazon narrative text from an earlier example, so read the numbers there, not the words.

Why it matters for the calculator: the golden test is this exact bridge, to six decimals.

Source: bkdrkpfnng.txt, sections 2.7 and 2.8; fcffsimpleginzu.md, "Valuation output", "Stories to Numbers" and "Diagnostics" step 6 (2026-02-01);
financeMD/damodaran/blog/2018/07/share-count-confusion-dilution-employee.md (2018-07-25)

The nine default assumptions you can override

OPTIONAL - FIRST PASS

Implementation reference. After you understand the nine stations, return here to inspect the switches and exact overrides.

Each override is a Yes/No switch plus a value cell that is read only when the switch says Yes.

#	Switch	What happens if No	What you can override
1	B45 stable cost of capital	Terminal rate = risk-free + mature premium (8.81%)	B46, your own terminal rate
2	B48 stable return on capital	Stable ROC = year-10 cost of capital	B49, a higher or lower ROC
3	B51 probability of failure	0%	B52 probability; B53 and B54 always read
4	B56 reinvestment lag	Lag = 1 year	B57, lag of 0 to 3
5	B59 tax convergence	Effective fades to marginal over years 6 to 10	Yes keeps the effective rate forever
6	B61 NOL carryforward	0	B62, opening loss bank
7	B64 risk-free after year 10	Today's risk-free forever	B65, a different long-run rate
8	B67 growth in perpetuity	g = risk-free (or B65 if #7 is Yes)	B68, your own g, negative allowed
9	B70 trapped cash	Cash added at face	B71 amount and B72 foreign tax rate

Why it matters for the calculator: every override is an optional field with a documented default, and override 2 changes terminal reinvestment, so it changes the terminal value.

Source: bkdrkpfng.txt, section 1.6 (2026-09)

The Diagnostics sheet checks

The Diagnostics sheet puts your forecast beside history and industry data, then asks questions. It never blocks anything. Almarai's numbers:

- Step 1, revenue growth: industry average 7.16%, most recent year 7.63%, forecast 5% for years 1 to 5. The question asked is why the forecast differs from recent history.
- Step 2, revenue in currency: 21,765.4 today, 35,030.0 in year 10. The questions are about the size of the total market and the implied share.
- Step 3, margins: 14.063% flat. The questions are about industry margins and unit economics.
- Step 4, reinvestment: sales to capital 0.59 most recent versus 1.71 in the forecast. The reinvestment drag is shown too: present value of after-tax operating income over ten years is 21,877.6, of which 5,483.1 (25.06%) is consumed by reinvestment, leaving 16,394.5 of FCFF.
- Step 4, return on capital ladder: 6.875% most recent, 14.50% marginal over years 1 to 10, 8.42% in year 10, 8.81% stable. Marginal ROC is the extra after-tax income divided by the extra capital, and it is the "is the growth worth it" test.
- Step 5, risk: industry cost of capital 5.79%, compounded 7.58%, years 1 to 5 7.055%, stable 8.81%. Failure rate 0.
- Step 6, value versus price: 0.0994, with a verdict string when the ratio is above 2 or below 0.5, and a fix-it matrix naming which input to move.

Why it matters for the calculator: the sheet has no hard validations, only dropdowns and text. The plan turns these checks into real errors and warnings, so the calculator should compute every number above.

Source: fcffsimpleginzu.md, "Diagnostics" (2026-02-01); bkdrkpfng.txt, section 4; plans/help-me-out-here-abstract-wilkinson.md, milestone 2 (2026-09)

Simple ginzu versus full ginzu

The simple ginzu treats revenue growth as the primitive. Margin turns revenue into operating income, and sales to capital turns revenue growth into reinvestment, over a fixed ten-year horizon with a built-in cost of capital worksheet and a failure module. The full ginzu (fcffginzu) is a different model family. Growth in operating income comes from fundamentals, meaning return on capital times reinvestment rate. Capital spending, depreciation and working capital are explicit lines. The high-growth period length is a user choice, beta and debt ratios are set separately for the growth and stable phases, an earnings normalizer sheet handles odd base years, and non-traded companies are supported. It has no failure probability module. The calculator ports the simple ginzu; the full one is only needed for fundamentals-driven growth or private companies.

Source: bkdrkpfng.txt, section 5 (2026-09)

Where this comes from

- financeMD/damodaran/pc/fcffsimpleginzu.md, the workbook itself; read “Valuation output”, “Stories to Numbers” and “Diagnostics” together.
- claude/tool-results/bkdrkpfng.txt, the formula-by-formula schema, including the nine overrides and the simple-versus-full table.
- financeMD/valpacket1spr25.md, slides 191 to 211 (growth, sales to capital, terminal value and stable growth) and slide 310 (distress).
- financeMD/damodaran/blog/2016/11/myth-55-terminal-value-ate-my-dcf.md, why a large terminal share is normal.
- financeMD/damodaran/blog/2016/11/myth-53-growth-is-good-more-growth-is.md, growth must be paid for with reinvestment.
- financeMD/damodaran/blog/2015/02/dcf-myth-1-if-you-have-ddiscount-rate.md, what a DCF is and is not.
- financeMD/damodaran/blog/2015/08/dcf-myth-2-dcf-is-exercise-in-modeling.md, why the numbers need a story.
- financeMD/damodaran/blog/2018/07/share-count-confusion-dilution-employee.md, options and share counts in the equity bridge.
- financeMD/damodaran/New_Home_Page/definitions.md, one-line definitions of FCFF, reinvestment rate and fundamental growth.

Three things to remember

KEY IDEA

1. The chain is revenue, margin, tax, reinvestment, FCFF, discount, terminal value, failure, bridge. Nine stations, in that order, and station 4 needs station 1 finished one year ahead.
2. **Growth is not free.** Every riyal of extra revenue costs one divided by the sales to capital ratio, and in the terminal year the cost becomes g divided by ROC.
3. Three terminal defaults do most of the work: growth equals the risk-free rate, stable ROC equals the cost of capital, and the terminal cost of capital equals risk-free plus the mature-market premium. Code the formulas, not the labels.

Chapter 4: Discount Rates and Country Risk

READ FIRST

Understand the parts of the discount rate. Start with the map, then the risk-free rate, equity premium and country risk. Pay particular attention to matching dates and currency.

This chapter is about one number: the cost of capital. Chapter 3 showed where it sits in the DCF chain and how it glides toward a terminal rate. This chapter opens the number up and shows the parts inside it. Each part has a source table, a formula in the workbook, and a trap. Every concept ends with one line on why it matters for the calculator.

The map in one block

The cost of capital is the **blended return** that lenders and owners together demand from a company. Damodaran calls it a hurdle rate, because a project must clear it to add value. It mixes a cost of equity and a cost of debt by market-value weights.

Cost of equity = risk-free rate + levered beta x equity risk premium

Pre-tax cost of debt = risk-free rate + company default spread + country default spread

After-tax cost of debt = pre-tax cost of debt x (1 - marginal tax rate)

Cost of capital = $E/(D+E) \times \text{cost of equity} + D/(D+E) \times \text{after-tax cost of debt}$

Every symbol in that block is a section below. The risk-free rate appears twice, so an error there travels everywhere.

Source: [financeMD/damodaran/blog/2026/02/data-update-5-for-2026-risk-and-hurdle.md](https://financeMD.com/damodaran/blog/2026/02/data-update-5-for-2026-risk-and-hurdle.md) (2026-02); bkdrkpfng.txt section 2.6

The risk-free rate

WATCH OUT

Match the currency and inspect default risk. This section explains why a government bond yield may need an adjustment before it is used here.

A risk-free rate is the return on an investment whose payoff you know for certain. In practice it is the yield on a long-term government bond. Damodaran's rule is that the bond must be in the **currency of your cash flows**, and the government must be close to default-free. A currency is a unit of measurement, like meters versus feet. Pick one, and the risk-free rate, the growth rate and the cash flows all live in it.

Source: [financeMD/damodaran/pdfiles/country/val2dayIndia2025.md](https://financeMD.com/damodaran/pdfiles/country/val2dayIndia2025.md), slides 28 to 30 (2025); bkdrkpfng.txt section 6

Stripping out the default spread

Most governments are not default-free. A default spread is the extra yield a borrower pays because it might not repay. Damodaran's fix is to subtract the country's default spread from the bond yield. His India session shows a 2025 example: an Indian bond yield of 6.32%, a default spread of 2.16% at the Baa3 rating, so a rupee risk-free rate of 4.16%.

Source: [financeMD/damodaran/pdfiles/country/val2dayIndia2025.md](https://financeMD.com/damodaran/pdfiles/country/val2dayIndia2025.md), slide 29 (2025)

The 2025 US downgrade and the 0.22%

On May 16, 2025, Moody's lowered the United States from Aaa to Aa1. It had been the last agency still rating the US at the top grade. By his own rule the treasury rate is no longer a pure risk-free rate. So his calculators now show two versions: the conventional one on the treasury rate, and an adjusted one that nets out the Aa1 default spread. The September 2026 calculator carries that spread as 0.22%, with a treasury rate of 4.75% and an adjusted dollar risk-free rate of 4.53%. His July 2025 India slide used 0.27%, so the spread itself moves from vintage to vintage.

Source: [financeMD/damodaran/pc/implprem/ERPSept26.md](#), "Impl premium calculator" sheet (2026-09); [financeMD/damodaran/pdfiles/country/val2dayIndia2025.md](#), slide 29 (2025)

Three rates in one year

The 2026 files hold three US risk-free rates: 3.95%, 4.58% and 4.75%. Each belongs to a different dated table, called a vintage. The workbook hard-codes 4.58% inside two formulas on the cost-of-capital sheet. That is why every run in our system will store three vintage ids: the market file, the country-risk file and the industry file.

Source: [Valuation/docs/read-this-first.md](#) section 3.2 (2026-09); [bkdrkpfnfng.txt](#) section 7, note 3; [bafb6h9rb.txt](#)

The India rule and a cross-check

For an Indian company valued in rupees, the risk-free rate is the 10-year government bond yield minus India's default spread. Otherwise sovereign risk is counted twice, because the cost of debt adds that spread back later. When you distrust the bond yield, take the dollar risk-free rate and add the expected inflation gap. His February 2026 post uses IMF forecasts of 2.24% for the US and 4.00% for India, a gap of 1.76%. If the two routes disagree by a lot, an input is stale.

Source: [Valuation/docs/read-this-first.md](#) section 3.4 (2026-09); [financeMD/damodaran/blog/2026/02/data-update-5-for-2026-risk-and-hurdle.md](#) (2026-02)

Why it matters for the calculator: the risk-free rate is an input with a currency and a vintage attached, never a bare number, and the India module subtracts the default spread before anything else runs.

The equity risk premium

The equity risk premium, or ERP, is the extra annual return investors demand for owning the stock market instead of the risk-free bond. It is the price of risk for the whole market. There are two ways to measure it, and Damodaran spent a career arguing for the second.

Historical versus implied

The historical premium is the average past gap between stock returns and bond returns. At the start of 2026 it ranged from 5.5% to 14.5%, depending on the period, the averaging method and the risk-free choice. He lists three problems. It looks backward. It is noisy, with large standard errors. And it moves the wrong way, falling after a crash because the bad year drags the average down.

Source: [financeMD/damodaran/blog/2026/03/the-price-of-risk-equity-risk-premium.md](#) (2026-03)

The implied premium looks forward. Think of a bond: you know its price and its coupons, so you can solve for the yield. He does the same for the S&P 500. Take the index level, the cash returned through dividends and buybacks, and an expected growth rate. Solve for the discount rate that makes the

present value equal the index level. That rate is the expected return on stocks. Subtract the risk-free rate, and the remainder is the implied ERP. In Excel he uses Goal Seek, which is a root finder.

Source: [financeMD/damodaran/pc/implprem/ERPSept26.md](#), “Impl premium calculator” sheet (2026-09)

The September 2026 numbers

Input or output, start of September 2026	Value
S&P 500 level	7686.14
Cash yield (dividends plus buybacks, trailing 12 months)	2.63%
Expected earnings growth, next five years	13.99%
10-year treasury rate	4.75%
Implied expected return on stocks	8.84%
Implied ERP over the treasury rate	4.09%
Adjusted dollar risk-free rate (4.75% minus 0.22%)	4.53%
Implied ERP over the adjusted rate	4.31%

For scale, the same sheet lists the historical US premium at 5.61%, the average implied premium of the last decade at 5.08%, and the average since 1960 at 4.25%.

Source: [financeMD/damodaran/pc/implprem/ERPSept26.md](#) (2026-09);
[financeMD/damodaran/pc/implprem/ERPbymonth.md](#), row 2026-09-01 (2026-09)

What “mature market premium” means after the downgrade

Until 2025 the US implied ERP was his mature-market premium, because the US was Aaa. Now he starts from the US premium and removes the piece that belongs to US sovereign risk. His country-risk file states the January 2026 arithmetic as 4.61% minus 0.23% times 1.52, which gives 4.23%. Two honest footnotes. That subtraction does not compute exactly (4.61% minus 0.35% is 4.26%). And his ERP spreadsheet for the same month shows 4.23% as the S&P 500 premium over the unadjusted T-bond of 4.18%, with 4.46% over the adjusted 3.95%. His files disagree on the derivation, not on the number the workbook uses. Code the workbook’s 4.23%, and store which file it came from. The July 2026 update starts at 4.42%, nets out 0.22%, and lands on 4.20%. The workbook keeps the mature premium in one cell, and that cell drives every country premium and the terminal cost of capital.

Source: [financeMD/damodaran/pc/datasets/ctryprem.md](#), “Explanation” rows (2026-01);
[financeMD/damodaran/blog/2026/07/country-risk-drivers-measures-and.md](#) (2026-07); [bkdrkpfng.txt](#) section 3.2

Chapter 6 covers the other use of this number, reading what the market is pricing in.

Why it matters for the calculator: the ERP is refreshed monthly and the mature premium quarterly, so the engine will read both from the vintage tables and never keep either as a constant in code.

The country risk premium

A country risk premium, or CRP, is the extra premium for equity in a riskier country, added on top of the mature-market premium. Damodaran builds it in three steps.

1. Start with the sovereign rating. A rating is an agency’s letter grade for a government’s ability to repay, such as Baa3 for India.

2. Convert the rating to a default spread using his ratings-to-spreads table. A CDS spread, which is the price of insuring the bond, is a market-based alternative that exists for only 84 countries.
3. Scale the spread up, because stocks are more volatile than government bonds. The scalar is the volatility of an emerging-market equity index divided by the volatility of an emerging-market government bond index. It was 1.52 in January 2026 and 1.55 in July 2026.

The total ERP for a country is the mature premium plus its CRP. His India teaching example: a 2% default spread, Sensex volatility of 21%, bond volatility of 14%, so a CRP of 3.00%.

Source: [financeMD/damodaran/blog/2026/07/country-risk-drivers-measures-and.md](#) (2026-07);
[financeMD/damodaran/pc/datasets/ctryprem.md](#) (2026-01);
[financeMD/damodaran/pdfiles/country/val2dayIndia2025.md](#), slide 45 (2025)

India, two vintages

Item	January 2026 vintage	July 2026 vintage
Moody's rating	Baa3	Baa3
Adjusted default spread	1.87%	1.75%
Mature-market ERP	4.23%	4.20%
Country risk premium	2.85%	2.72%
Total ERP for India	7.08%	6.92%

Same rating, different numbers, six months apart. The spread table moved and the scalar moved. A run that takes the July ERP with the January industry table has mixed two vintages without noticing.

Source: [bafb6h9rb.txt](#) section 1 ([ctryprem.xlsx](#) and [ctrypremJuly26.xlsx](#));
[financeMD/damodaran/pc/datasets/ctryprem.md](#), India row (2026-01)

Where the company operates, not where it is registered

He argues that exposure comes from where a company earns, not where it is registered. The workbook offers four ERP choices: type it in, use the country of incorporation, weight by operating countries, or weight by operating regions. The weights are revenue shares. He notes revenues suit consumer companies, production suits natural-resource companies, and a mix suits manufacturers. Almarai, the golden case, uses region weights and gets a blended ERP of 5.58% rather than the Saudi premium.

Source: [financeMD/damodaran/blog/2026/07/country-risk-drivers-measures-and.md](#) (2026-07);
[bkdrkpfng.txt](#) section 2.6; [financeMD/damodaran/pc/fcfsimpleginzu.md](#), "Cost of capital worksheet" (2026-02)

Why it matters for the calculator: the country table is a dated file of about 200 rows looked up by name, and the revenue-weighted option needs a segment table the memo writer will extract from the filing.

Betas

A beta measures how much a stock moves when the whole market moves. A beta of 1 moves with the market, 0.5 moves half as much. In the cost-of-equity formula it scales the ERP for one company.

Regression betas are noisy

The textbook way is to regress the stock's past returns on the market's past returns. Damodaran's verdict is blunt: regression betas “will almost always be either too noisy or skewed by estimation choices”. The standard error, the statistical uncertainty around the slope, can be as large as the estimate. The result also shifts with the index, the period and the return interval you picked.

Source: “financeMD/Aswath Damodaran - Investment Valuation, University Edition _ Tools and Techniques for Determining the Value of Any Asset (2023).md”, chapter 8, “Bottom-Up Betas” (2023)

Bottom-up betas

The fix is averaging. Find the companies in the same business, average their regression betas, and strip out their debt to get an unlevered beta. An unlevered beta is the beta the business would have with no debt. If a company is in several businesses, weight the business betas by value, or by revenue if value is unavailable. Then add back the company's own debt. The arithmetic: if each firm's beta has a standard error of 0.50 and there are 100 firms, the average has a standard error of 0.05.

Source: same book, chapter 8, “The Case for Bottom-Up Betas” (2023)

Unlevering, relevering, and cash

Debt makes equity riskier, so a beta must be adjusted for how much debt sits under it. The formula, usually called the Hamada equation, is the one the workbook uses.

Levered beta = unlevered beta $\times$ (1 + (1 – tax rate) $\times$ Debt/Equity)

Unlevered beta = levered beta / (1 + (1 – tax rate) $\times$ Debt/Equity)

Debt and equity are at market value. Cash has a beta near zero, so a company with a lot of cash looks safer than its business is. His industry tables carry an “unlevered beta corrected for cash” column, which divides by one minus cash as a share of firm value. That is the column the workbook reads.

For Almarai the workbook takes the US industry unlevered beta of 0.469, a market debt-to-equity ratio of 39,954 over 311,888, and a 25% tax rate. That produces a levered beta of 0.515.

Source: bkdrkpfn.txt section 2.6; financeMD/damodaran/pc/datasets/betaIndia.md, “Explanation” rows (2026-01); financeMD/damodaran/pc/fcfsimpleginzu.md, “Cost of capital worksheet” (2026-02)

Industry averages and the India fallback

His industry betas average each firm's 2-year and 5-year weekly regression betas, with the 2-year beta weighted two thirds. The workbook ships only US and Global tables of 94 industries each. India has its own table over the same 94 industries, built from 5,170 companies, but it is thin in places. Software (System and Application) has 82 firms and a cash-corrected unlevered beta of 0.91. Air Transport has 5 firms. Utility (General) has none. When an Indian industry has fewer than 10 firms, true for 27 of the 94, the engine will use the emerging-markets table, then Global, and print which one it used.

Source: financeMD/damodaran/pc/datasets/betaIndia.md (2026-01-05); bafb6h9rb.txt sections 3 and 4; Valuation/docs/read-this-first.md section 3.4 (2026-09)

Why it matters for the calculator: beta will come from a lookup keyed on industry name and vintage, relevered with the company's own market debt-to-equity, and the output will state the tier that supplied it.

The cost of debt

The cost of debt is the rate the company would pay today to borrow long-term. Not the coupon on old bonds, and not interest expense divided by book debt. It is the risk-free rate plus the company's default spread plus the country's default spread.

Synthetic ratings from interest coverage

Most companies have no bond rating. Damodaran plays the rating agency himself with one ratio: interest coverage, which is operating income divided by interest expense. Higher coverage means interest is paid more comfortably. He maps coverage bands to ratings and spreads in three tables: large non-financial firms, smaller and riskier firms, and financial-service firms. The table choice is the "type of firm" input.

Almarai's coverage is 3,060.9 over 493.4, or 6.20. In the smaller-firm table that is A2/A with a 0.7751% spread. Add the Saudi default spread of 0.51% and the 4.58% risk-free rate, and the synthetic cost of debt is 5.8651%. The workbook's chosen route for Almarai is an actual rating of A1/A+, spread 0.7003%, giving 5.2803%. The actual-rating formula adds no country spread. The golden test must match that behavior, and the audit will flag it.

Source: bkdrkpfng.txt sections 2.6 and 3.4; financeMD/damodaran/pc/fcffsimpleginzu.md, "Synthetic rating" sheet (2026-02)

Two special cases

Negative operating income sends coverage below every band and returns a D rating. The sheet warns not to use that as a going-concern cost of debt and suggests BB instead. Missing interest expense is the opposite trap. The formula sets coverage to one million when interest is zero, which awards Aaa. That is harmless for a company with no debt, because debt then has no weight. It is wrong for a company with debt that did not break out interest in the filing. The plan lists "interest missing" as an audit field, so the engine will record the gap instead of quietly rewarding it.

Source: bkdrkpfng.txt section 3.4; plans/help-me-out-here-abstract-wilkinson.md section 10.2 (2026-09)

Why it matters for the calculator: the rating module is three banded tables plus a country lookup, and the 2026 spreads run from 0.40% at Aaa to 19% at D.

Weights, and the four approaches

Weights must be **market values, not book values**. Book value is the accounting record of what was raised long ago. Market value is what the claims are worth today, which is what a discount rate is about. Equity is shares times price. Most debt does not trade, so the workbook treats book debt as one bond: interest expense is an annuity over the average maturity, book value is the final repayment, both discounted at the pre-tax cost of debt.

Almarai's pieces come together like this. Cost of equity is 4.58% plus 0.515 times 5.58%, or 7.45%. After-tax cost of debt is 5.2803% times 0.75, or 3.96%. Market equity is 311,888 and market debt is 39,954, so the weights are 88.6% and 11.4%. The cost of capital is 7.055%, the rate Chapter 3 carried through years 1 to 5.

Source: financeMD/damodaran/pc/fcffsimpleginzu.md, "Cost of capital worksheet" (2026-02); bkdrkpfng.txt section 2.6

The worksheet lets you reach that number four ways.

Approach	What it does	When he uses it
I will input	You type the cost of capital	Reverse engineering, or a quick pass
Detailed	Builds it from the parts above	The default for a real valuation
Industry Average	Reads the industry cost of capital from the US or Global table	Young firms, or as a cross-check
Distribution	Picks a decile (a cut point that splits all firms into tenths) or quartile (quarters) from the histogram of all firms in a region	Firms whose risk is in flux

The last two carry a hidden conversion. The industry tables are in US dollars at the vintage's risk-free rate. The sheet adds your risk-free rate minus 4.58% to bring the figure up to date and into your currency. The India cost-of-capital table also publishes a local-currency column: 10.92% for the whole market against 8.28% in dollars, and 12.67% against 9.99% for Software (System and Application).

Source: [bkdrkpfng.txt](#) section 2.6; [financeMD/damodaran/pc/datasets/wacclIndia.md](#) (2026-01-05)

Why it matters for the calculator: the engine will store each table's embedded risk-free rate next to the table, so the conversion is "your rate minus the table's rate", never a literal 4.58%.

Plausibility bands

His most-used table is the histogram of costs of capital for all 48,156 firms in his sample. He uses it as a starting input for young firms, as a plausibility check on other people's work, and as a reminder not to spend the day polishing the fourth decimal.

Region, start of 2026, US dollars	First decile	Median	Ninth decile
US	5.26%	7.79%	9.88%
Global	6.28%	8.65%	11.66%
Emerging markets	6.65%	8.95%	12.55%

His rule of thumb: 80% of US companies sit between 5.26% and 9.88%, and 80% of global companies between 6.28% and 11.66%. A dollar cost of capital of 14% for a listed US company "goes on my suspect list". A small, newly listed AI firm might start at 11.66% and drift toward 8.65%. For rupee valuations, add the 1.76% inflation gap to every band first.

Source: [financeMD/damodaran/blog/2026/02/data-update-5-for-2026-risk-and-hurdle.md](#) (2026-02); [financeMD/damodaran/pc/fcfsimpleginzu.md](#), "Cost of capital worksheet", distribution table (2026-02)

Why it matters for the calculator: the diagnostics step will compare each run's cost of capital against these bands, in the right currency, and raise a warning outside the first-to-ninth decile range.

Where this comes from

Go deeper in this order. Paths are under [financeMD/](#) unless shown in full.

- [damodaran/blog/2026/02/data-update-5-for-2026-risk-and-hurdle.md](#): risk measures, hurdle rates, and the 2026 cost-of-capital histogram with the plausibility bands.

- [damodaran/blog/2026/03/the-price-of-risk-equity-risk-premium.md](#): historical versus implied ERP, the 2008 daily series, and the three lessons.
- [damodaran/blog/2026/07/country-risk-drivers-measures-and.md](#): the full country-risk method, the 1.55 scalar, and the case for operating exposure over incorporation.
- [damodaran/pc/implprem/ERPSept26.md](#) and [ERPbymonth.md](#): the implied-ERP calculator and the monthly series the regime layer will load.
- [damodaran/pc/datasets/ctryprem.md](#) and [ctrypremJuly26.md](#): the country tables, the “Explanation” rows on the mature premium, and the relative volatility sheet.
- [damodaran/pc/datasets/betalIndia.md](#) and [wacclIndia.md](#): the India industry tables with firm counts, cash-corrected betas, and the local-currency column.
- [damodaran/pdfiles/country/val2dayIndia2025.md](#), slides 28 to 52: risk-free rates in rupees, the CRP arithmetic, and lambda, taught in India.
- The 2023 book at the corpus root, chapters 7 and 8: standard errors on historical premiums, and the case for bottom-up betas.
- [damodaran/pc/fcfsimpleginzu.md](#), the “Cost of capital worksheet”, “Synthetic rating” and “Country equity risk premiums” sheets: the formulas the engine must reproduce.

Three things to remember

KEY IDEA

1. The cost of capital is five inputs, and four of them come from dated tables: the risk-free rate, the mature ERP, the country spread and the industry beta. A run without its three vintage ids is not a run.
2. Strip default risk out of the risk-free rate and add it back only in the cost of debt and the country premium. Counting it twice is the most common India error.
3. Do not polish the number; check it. Bottom-up betas beat regressions, the implied ERP beats history, and a result outside his deciles is a signal to reread the inputs, not a discovery.

Chapter 5: Growth, Terminal Value, and the Dark Side

READ FIRST

Read growth, terminal value, and ranges carefully. These sections explain why an exact spreadsheet answer can still be highly sensitive to the assumptions.

What this chapter is

Chapter 3 walked through the workbook’s arithmetic station by station. This chapter covers the two places where that arithmetic is most easily abused: growth and terminal value. Then it covers the companies where the standard recipe breaks, which Damodaran calls the “dark side” of valuation. Every rule below exists because someone, often Damodaran himself, has shown how easy it is to make a spreadsheet say anything.

Growth is bought, not given

The growth equation

Growth in operating income comes from two things: how much a company reinvests, and what return it earns on that reinvestment. Reinvestment rate is the share of after-tax operating income put back into the business. Return on capital is after-tax operating income divided by the capital already invested. Multiply them and you get expected growth. The packet's worked example: a reinvestment rate of 83.33% times a return of 12% gives 10% growth. Think of a bakery. To bake more bread next year you must add an oven this year. The oven must then produce bread worth more than it cost.

Source: financeMD/valpacket1spr25.md, slides near lines 2251 and 2407 (Spring 2025)

Why it matters for the calculator: the simple ginzu asks for revenue growth and a sales-to-capital ratio, and derives reinvestment from them. The full ginzu goes the other way, growth from return times reinvestment. Same equation, read in different directions.

Source: bkdrkpfnng.txt, section 2.5 and the full-ginzu comparison note

Sales to capital, the efficiency lever

Sales to capital is revenue divided by invested capital. It answers: how much revenue does one unit of invested capital produce? A high ratio makes growth cheap. A low ratio means each extra unit of revenue needs a lot of new capital. In the simple ginzu, reinvestment each year equals the change in revenue divided by this ratio. So it is the single lever that decides how much of the growth story the company must pay for. Chapter 3, station 4, has the exact formula and the lag switch.

Source: bkdrkpfnng.txt, section 2.5; fcffsimpleginzu.md, "Valuation output" row 8 (2026-02-01)

Why it matters for the calculator: the workbook shows the company's ratio next to the US industry, the global industry, and the global quartiles. A ratio above the industry's third quartile needs a written justification before it enters a base case.

Source: bkdrkpfnng.txt, section 1.7 (feedback block, rows 25 to 31)

Growth that destroys value

Growth is not automatically good. If the return on new capital is below the cost of capital, every unit reinvested is worth less than it cost, and growth shrinks value. The packet's table shows this. When return on capital equals cost of capital, terminal value sits at 1,000 whatever the growth rate. With a 6% return, raising growth from 0% to 3% drops it to 714. With a 14% return, the same change lifts it to 1,122. In his words, "It is not growth per se that creates value but growth with excess returns."

Source: financeMD/valpacket1spr25.md, slides 206 to 208 (Spring 2025);
financeMD/damodaran/blog/2016/11/myth-53-growth-is-good-more-growth-is.md (2016-11-30)

Why it matters for the calculator: the year-by-year return on invested capital row exists so you can see whether the story's growth earns its keep. Revenue tripling while ROIC stays below the cost of capital should print a warning.

Source: bkdrkpfnng.txt, section 2.5 (rows 39 and 40)

All good things end

How long can high growth last

Most companies cannot grow fast for long, and almost none can do it while earning excess returns, meaning returns above the cost of capital. Damodaran's four propositions: a larger potential market

allows longer growth; a smaller company relative to that market allows longer growth; stronger and more durable competitive advantages allow longer value-creating growth; and such advantages are rare. His practical cap: a growth period much longer than ten years means “you are already building in the expectation that your firm is an exceptional firm.”

Source: financeMD/valpacket1spr25.md, slides 206 and 294 (Spring 2025)

Why it matters for the calculator: the simple ginzu fixes the high-growth window at ten years. That is a design choice. Do not extend the grid.

Source: bkdrkpfng.txt, comparison table row “Horizon”

Convergence: what fades and what it fades to

By the terminal year a company must look like a stable company, so several things fade at once. Revenue growth falls to the stable rate. Operating margin converges to a target. The tax rate converges to the marginal rate. Cost of capital converges to that of a mature company. Return on capital converges to cost of capital, so excess returns approach zero. Chapter 3 shows how the workbook phases each over years 6 to 10. Damodaran notes the McKinsey view that return on capital should always equal cost of capital in stable growth. He adds his own reservation: “excess returns seem to persist for very long time periods.” The workbook defaults to full convergence but lets you override the stable return on capital.

Source: financeMD/valpacket1spr25.md, slides 209 to 210 (Spring 2025); bkdrkpfng.txt, section 2.5 (ROC_stable)

Why it matters for the calculator: every override of a convergence default is a claim that the company is unusual. Log the override and its justifying sentence together in the run record.

Terminal value discipline

WATCH OUT

Keep all three rules together. Growth, return on capital, and reinvestment must tell a consistent long-run story.

The three rules

Terminal value is one number standing in for every cash flow after year ten. Chapter 3, station 7, has the formula and the Almarai numbers. Three rules keep it honest.

Rule	What it says	Why
Growth cap	Stable growth at or below the risk-free rate, in the currency of the cash flows	The risk-free rate is a proxy for nominal growth in the economy, and no company outgrows its economy forever
Excess returns fade	Stable return on capital equals cost of capital by default	Competition erodes advantages; assuming otherwise is a claim you must defend
Consistent reinvestment	Terminal reinvestment rate equals stable growth divided by stable return on capital	Growth forever still has to be paid for forever

On the growth cap, Damodaran does not forecast the economy. He uses the risk-free rate because it is observable, carries the currency's inflation automatically, and tracks nominal growth. In the US from 1954 to 2015 nominal GDP growth ran about 0.74% above the risk-free rate, and since 1981 it has lagged it by about 0.58%. His conclusion: stable growth below the economy's rate cannot hurt you, above it can make valuations implode. Negative stable growth is allowed and sometimes right for declining businesses.

Source: financeMD/damodaran/blog/2016/11/myth-52-as-g-rto-infinity-and-beyond.md (2016-11-30); financeMD/damodaran/blog/2016/11/myth-54-negative-growth-rates-forever.md (2016-11-30); valpacket1spr25.md, slides 201 to 205

On consistent reinvestment, the common mistake is to grow the year-ten free cash flow by one more year and call it the terminal cash flow. That silently locks the year-ten reinvestment rate in forever. The workbook instead recomputes terminal reinvestment as growth divided by return on capital. For Almarai that is 4.58% divided by 8.81%, a 51.99% rate, which is why reinvestment jumps in the terminal column.

Source: financeMD/damodaran/blog/2016/11/myth-53-growth-is-good-more-growth-is.md (2016-11-30); bkdrkpfng.txt, section 2.5 (Reinv_T)

Why it matters for the calculator: the terminal cost of capital is risk-free rate plus mature-market equity risk premium (ERP), whatever the sheet label says. Code the formula, not the label, or the golden test fails.

Source: bkdrkpfng.txt, section 1.6 note on the label mismatch

Why terminal value dominates, and what that means

In the Almarai golden case the present value of the terminal value is 21,122.99 and the present value of the ten explicit years is 16,394.54. So the terminal piece is about 56% of operating value. For growth companies it can be 80%, 90%, or more than 100%, because early cash flows are small or negative. Critics call this a flaw. Damodaran's answer has two parts. First, it is normal: most of the historical return on stocks came from price appreciation, not dividends, so most of a DCF's value should sit at the end. Second, a large terminal share means the growth-phase assumptions matter more, not less. Terminal value is computed from year-ten earnings, and those are whatever the first ten years produced.

Source: bkdrkpfng.txt, section 2.7 (B19 and B20); financeMD/damodaran/pdfiles/CLC/slides/Ch11.md, lines 105 to 109; financeMD/damodaran/blog/2016/11/myth-55-terminal-value-ate-my-dcf.md (2016-11-30)

Why it matters for the calculator: print the terminal share on every run. A share far above the industry norm is a prompt to reread the margin and growth story, not a reason to distrust the method.

The Dark Side of Valuation

What the book is

"The Dark Side of Valuation" began as a paper valuing Amazon in March 2000, when his DCF gave about 34 dollars a share against a price of 80. The first edition covered young technology companies. The second, after 2008, added distressed companies, commodity firms, and banks. The third edition is built around three macro problems: very low or negative interest rates, risk premiums that move violently and often, and rising political risk as globalization reverses. It has twenty chapters in five parts and ends with "Lighting the Way: Vanquishing the Dark Side." The thesis: the dark side is not a different model. It is the same model applied where data is thin, and the temptation is to abandon it

for “paradigm shifts,” invented metrics, and stories that outrun numbers. Refusing to value something because it is uncertain “is a cop-out, since not dealing with uncertainty does not make it go away.”

Source: financeMD/damodaran/New_Home_Page/darkside/preface.md;
financeMD/damodaran/New_Home_Page/DSV3/dsv3edpreface.md; valpacket1spr25.md, slide 291;
financeMD/damodaran/blog/2014/06/a-disruptive-cab-ride-to-riches-uber.md (2014-06-09); bkm2kob5x.txt, Part 1

What changes, by company type

The plan requires every memo to name the company’s uncertainty type and pull the matching passages. Here is the map.

Type	What breaks	What Damodaran changes	Calculator hook
Young or growth	Little history, small revenue, losses, real chance of failure	Work top down from market size and share; let cost of capital fall over time; keep failure risk out of the discount rate; expect share count to rise	Failure probability on; cost-of-capital path from high to mature; option and dilution inputs
Distressed or declining	DCF assumes survival to the terminal year	Blend going-concern value with distress-sale value; allow negative growth and negative reinvestment	Probability of failure and proceeds rule; negative stable growth allowed
Cyclical or commodity	Base-year earnings are a point on a cycle, not a level	Normalize earnings across the cycle, or forecast the commodity price and say so; keep macro views separate from company views	Base year flagged as normalized, with the method stored
Financial service	Debt is raw material, not capital; cash flows are hard to define; regulators set capital ratios	Value equity directly, often with a dividend model; treat book equity as meaningful	The free-cash-flow-to-firm (FCFF) workbook is the wrong tool; the type label should raise a hard warning
Complex holding structures	Consolidated statements mix parent and holdings; cross holdings are opaque	Value the parent alone, then each holding, then add and subtract	Non-operating assets and minority interests as separate, cited rows
Emerging market	Country risk, currency and inflation shifts, governance, nationalization, more cross holdings	Add a country risk premium scaled from default spreads; strip the sovereign spread from the local risk-free rate	Country-risk vintage id on every run; INR risk-free equals the 10-year Indian government bond (G-sec) yield minus India’s default spread

Source: financeMD/valpacket1spr25.md, slides 292 to 321 and 339 (Spring 2025);
financeMD/damodaran/pdfiles/CLC/slides/Ch13.md; bafb6h9rb.txt (India rows);
plans/help-me-out-here-abstract-wilkinson.md, memo section

For India, the January 2026 vintage carries a default spread of 1.87% and a country risk premium of 2.85% on top of the mature premium of 4.23%. That is the concrete form “emerging market” takes in the numbers.

Source: bafb6h9rb.txt (India, January 2026 vintage)

Failure probability and recovery

A DCF “values a firm as a going concern.” If the firm may die before reaching stable growth, the DCF overstates value. The fix is one blend: going-concern value times one minus the probability of failure, plus distress-sale value times that probability. Chapter 3, station 8, shows the cells. Three ways to estimate the probability: the bond rating and its cumulative default table, a statistical model, or the market price of the company’s bonds. The workbook’s Failure Rate worksheet holds the rating table. For example, the ten-year cumulative default probability is 11.78% for BB and 50.38% for CCC and below. It also holds US Bureau of Labor Statistics survival rates by company age. It has no formula links; you read a number and type it in. His own choices show the range: Tesla in September 2013 at 10% failure with 50% recovery, Tesla in March 2014 at 0% after its bond issue, Uber in June 2014 at 10%.

Source: financeMD/valpacket1spr25.md, slide 310 and the Uber narrative slide (Spring 2025); bkdrkpfng.txt, section 3.5; financeMD/damodaran/blog/2014/03/return-to-firing-line-revisiting-tesla.md (2014-03-25)

Why it matters for the calculator: the probability and the proceeds rule are inputs the memo must state, with the table row they came from. A blank here should block the run for any company typed young or distressed.

DCF is gameable

The sales-to-capital 10 example

The strongest sensitivity critique of DCF is Damodaran’s own. In March 2014 his Tesla valuation used a sales-to-capital ratio of 1.55 and gave a value between 110 and 115 dollars a share. Then he showed the trick. Set the ratio to 10, meaning ten dollars of revenue per dollar invested, and the value jumps to 302 dollars a share. His comment: “Magical, right?” The catch is that this assumes Tesla adds about 68 billion dollars of revenue and 9 billion dollars of profit over a decade without building factories. Tesla had just announced a 5 billion dollar battery plant. His line on fake DCFs: cash flows and a discount rate on a spreadsheet no more make a DCF than “wearing tights and ballet shoes makes you a ballerina.”

Source: financeMD/damodaran/blog/2014/03/return-to-firing-line-revisiting-tesla.md (2014-03-25); bkm2kob5x.txt, Part 1 bullet 9 and section 3.4

Published ranges

He publishes the gameability as a feature. In February 2020 his Tesla grid crossed three stories with three revenue scales and gave values from 106 dollars to 2,106 dollars a share. In the same post the median global auto company generated 1.37 dollars of revenue per dollar of capital, and the 75th percentile 2.42. His own estimate of 3 was, he wrote, “more aspirational than based on observable efficiencies.” A twenty-fold range from one company on one day is the honest picture of how much inputs matter.

Source: financeMD/damodaran/blog/2020/02/a-do-it-yourself-diy-valuation-of-tesla.md (2020-02-06); bkm2kob5x.txt, section 3.4

Why it matters for the calculator: the guard rail is not a smarter formula. It is a plausibility band on each input, taken from his cross-sectional data, plus a written justification for anything outside it.

Ranges instead of points

KEY IDEA

This is how the reader sees uncertainty. Follow the link between varying the inputs, the sensitivity table, and the range of results.

Monte Carlo over his industry quartiles

His answer to input uncertainty is to stop pretending you know the number. If a key variable is uncertain, “why not quantify the uncertainty in a distribution (rather than a single price) and use that distribution in your valuation.” That is a Monte Carlo simulation. You draw each uncertain input from a distribution many times, run the model each time, and look at the spread of values. The output is a value distribution, plus the probability that value sits below price. The plan commits the system to this from M2, with distributions from his industry quartiles, and the dashboard will show the ranges.

Source: [financeMD/valpacket1spr25.md](#), slide 340 (Spring 2025); the approved plan, M2 section; [Valuation/docs/read-this-first.md](#), section 3.5

The “Input Stat Distributioons” sheet

The workbook already carries the raw material, in a sheet whose name is misspelled exactly like that. It holds 94 global industry groups. For each it gives the first quartile, median, and third quartile of six inputs: revenue growth over the last three years, pre-tax operating margin, sales to invested capital, cost of capital, beta, and debt to capital. Almarai’s group, Food Processing, has 1,450 firms. Its sales-to-invested-capital quartiles are about 0.90, 1.54, and 2.47. Its cost-of-capital quartiles are about 6.62%, 7.18%, and 7.91%. The Input sheet’s feedback block reads this table to show where the company sits against its peers.

Source: [financeMD/damodaran/pc/fcfsimpleginzu.md](#), “Input Stat Distributioons” sheet, Food Processing row (2026-02-01); [bkdrkpfng.txt](#), sections 1.7 and 3.3

Why it matters for the calculator: this lookup is an exact match on the global industry name, unlike the other industry tables. Load the sheet into its own table, keep the misspelling as the source label, and draw Monte Carlo inputs from it with the vintage id attached. A triangle across the three quartiles is a reasonable first distribution; that is a modeling choice you state, not a rule from the source.

The 2020 COVID series: the method under stress

The best test of a method is a crisis. From February to November 2020 Damodaran published fourteen posts while valuing through the crash. He kept computing the implied equity risk premium as prices fell; on April 1, 2020 it stood at 6.01%, up from 5.20% at the start of the year. He refused to switch to multiples and argued that prices have no bounds but “Value on the other hand has both upper and lower bounds,” set by cash flows, growth, and risk. In June he valued the S&P 500 with a simulation: the median value was 2,932 while the index traded near 3,100, around the 80th percentile of his distribution. He also lowered perpetual growth to match the lower risk-free rate, the growth cap doing its job in a low-rate world. The March 2020 snapshot of the simple ginzu, the corona variant, used a risk-free rate of 0.85% and had no failure-rate or distribution sheets yet.

Source: [financeMD/damodaran/blog/2020/03/a-viral-market-meltdown-iii-pricing-or.md](#) (2020-03-16); [financeMD/damodaran/blog/2020/04/a-viral-market-meltdown-vi-price-of-risk.md](#) (2020-04);

financeMD/damodaran/blog/2020/06/a-viral-market-update-ix-do-it-yourself.md (2020-06-04);
bkdrkpfnfng.txt, corona variant note; bkm2kob5x.txt, section 2.9

Why it matters for the calculator: a crisis changes the market vintage, not the engine. The ERP series, the risk-free rate, and the country spreads move; the formulas do not. That is why every run stores three vintage ids and why the regime panel is descriptive rather than predictive.

Source: bafb6h9rb.txt and the approved plan (three vintage ids per run)

Where this comes from

- financeMD/valpacket1spr25.md, slides 201 to 211 (terminal value), 206 (growth propositions), 291 to 321 (dark side by company type), 339 to 340 (macro views and simulation)
- financeMD/damodaran/New_Home_Page/DSV3/dsv3edpreface.md, the third edition's plan and chapter list
- financeMD/damodaran/New_Home_Page/darkside/preface.md, the original book's origin in the Amazon 2000 paper
- financeMD/damodaran/blog/2016/11/, DCF myths 5.1 to 5.5 on growth caps, growth and value, negative growth, and terminal value share
- financeMD/damodaran/blog/2014/03/return-to-firing-line-revisiting-tesla.md, the sales-to-capital 10 demonstration
- financeMD/damodaran/blog/2020/02/a-do-it-yourself-diy-valuation-of-tesla.md, the published grid of values and the auto-industry efficiency distribution
- financeMD/damodaran/blog/2020/, the fourteen "Viral Market" posts, especially III, VI, and IX
- financeMD/damodaran/pdfiles/CLC/slides/Ch11.md, Ch12.md and Ch13.md, terminal value at growth, mature, and declining firms
- financeMD/damodaran/pc/fcfsimpleginzu.md, the "Input Stat Distributions" and "Failure Rate worksheet" sheets
- Valuation/docs/other-side/, Document D, for the critics' side of terminal value and input sensitivity

Three things to remember

KEY IDEA

1. Growth is reinvestment times return, so every growth story has a bill. Sales to capital is where the bill is set, and a ratio above the industry's third quartile needs a sentence of defense before it enters a base case.
2. Terminal value follows three rules: growth at or below the risk-free rate, excess returns fading to zero by default, reinvestment equal to growth divided by return on capital. A large terminal share is normal and means the ten-year path matters more, not less.
3. The dark side is the same model with less data. Name the company's uncertainty type, state the failure probability and recovery rule, and report **a range** from his industry quartiles. A point estimate with no range hides the uncertainty; it does not remove it.

Chapter 6: Relative Valuation and What the Market Prices In

KEY IDEA

Keep pricing and valuing separate. Read the beginning of each part and the final takeaways, then return to the detailed multiple and market-premium examples.

Quick orientation before the dump. This chapter covers two things that look different but share one idea. First, pricing a single company against similar companies using multiples. Second, pricing the whole stock market using the implied equity risk premium. In both cases you are reading what the market is paying, not deciding what something is worth. Damodaran keeps those two activities apart, and so will your calculator. Corpus paths below are relative to financeMD/damodaran/ unless they start with a slash.

Part A: Pricing one company against its neighbors

What a multiple is

A multiple is a price divided by something that scales with the size of the business. Think of house prices per square foot. You cannot compare a large house with a small one on total price, but you can compare price per square foot. In stocks the numerator is the market price of equity or the enterprise value. Enterprise value means market equity plus debt minus cash, the price of the operating business. The denominator is earnings, book value, revenues, or EBITDA. EBITDA means earnings before interest, taxes, depreciation, and amortization, a rough operating cash flow before reinvestment. Damodaran's packet says standardizing prices "creates price multiples", and that "Almost 85% of equity research reports are based upon a multiple and comparable firms."

Source: pptfiles/val3E/valpacket2spr25.md, slides "The Essence of Relative Valuation" and "Relative valuation is pervasive" (Spring 2025)

His opening speaker note is blunt: "Most self proclaimed experts on valuation are really experts on pricing". Relative valuation is the formal name. Pricing is what it actually is.

Source: pptfiles/val3E/valpacket2spr25.md, title slide notes (Spring 2025)

Why it matters for the calculator: the industry averages sheet inside fcffsimpleginzu carries EV/EBITDA, EV/EBIT, Price/Book and Trailing PE, but the DCF never reads them. Only EV/Sales is used, and only to weight the parts of a multi-business company. Multiples belong in a separate pricing panel, never inside the value.

Source: claude/tool-results/bkdrkpfng.txt, industry table rows 15 to 19 (2026)

Pricing is not valuing

His closing proposition on the topic is one sentence: "Relative valuation is pricing, not valuation." A multiple tells you what the crowd is paying for similar assets today. It says nothing about whether the crowd is right. His warning: "Your relative valuation judgment can be right and your stock can be hopelessly over valued at the same time."

Source: pptfiles/val3E/valpacket2spr25.md, slide "Relative Valuation: Some closing propositions" (Spring 2025)

Why it matters for the calculator: every output screen labels a number as value (from the DCF chain) or price (from multiples or the market). The two never share a cell.

The four steps: define, describe, analyze, apply

He deconstructs any multiple in four steps. The table is the whole method in one place.

Step	The question	The trap it catches
Define	How exactly is this multiple computed?	Numerator and denominator belonging to different claimholders; trailing versus forward earnings
Describe	What does the distribution across all firms look like?	Judging a number as high or low with no sense of the range
Analyze	Which fundamentals drive it, and how?	Assuming the relationship is a straight line
Apply	Who is comparable, and how do I control for differences?	Treating “same sector” as “same fundamentals”

Source: pptfiles/val3E/valpacket2spr25.md, slides “The Four Steps to Deconstructing Multiples” and “Reviewing: The Four Steps” (Spring 2025)

Define. Proposition 1: “Both the value (the numerator) and the standardizing variable (the denominator) should be to the same claimholders in the firm.” Equity price goes over equity earnings or equity book value. Enterprise value goes over operating numbers such as EBITDA or sales. Price to EBITDA is inconsistent, and firms carrying more debt show a lower multiple for no good reason. He also lists the many faces of PE: current or average price; last financial year, trailing twelve months, or forecast earnings. In the 1990s bullish analysts used forward PE and bearish analysts used trailing PE, because the two disagreed.

Source: pptfiles/val3E/valpacket2spr25.md, slides “Definitional Tests” and “Example 1: Price Earnings Ratio” with notes (Spring 2025)

Describe. Multiples have skewed distributions, so “The averages are seldom good indicators of typical multiples.” Use medians and percentiles. Also check for bias: a PE cannot be computed for a loss-making firm, so a PE sample silently drops the weakest companies.

Source: pptfiles/val3E/valpacket2spr25.md, slides “Descriptive Tests”, “Multiples have skewed distributions” and “Reviewing: The Four Steps” (Spring 2025)

Why it matters for the calculator: store the definition next to every stored multiple (which price, which earnings, which date), and store the sample median and quartiles, never just the mean.

Analyze: the fundamentals hiding inside every multiple

Proposition 2 is the bridge between this chapter and the DCF chapters: “Embedded in every multiple are all of the variables that drive every discounted cash flow valuation - growth, risk and cash flow patterns.” Take a stable-growth dividend discount model and divide both sides by earnings. Out comes a PE formula with three inputs: expected growth, risk (the cost of equity), and how much can be paid out after reinvesting for growth. The same trick works for any multiple. Start from an equity model for equity multiples and a firm model for enterprise value multiples.

Source: pptfiles/val3E/valpacket2spr25.md, slides “Analytical Tests”, “A Simple Analytical device” and “I. PE Ratios” with notes (Spring 2025)

Each multiple has one driver that matters most. He calls it the companion variable: “Every multiple has one variable that can be considered its key determinant - its companion variable”.

Multiple	Consistent pair	Companion variable	Other drivers
PE	Equity price over equity earnings	Expected growth in earnings	Cost of equity, payout ratio, quality of growth
Price to book (PBV)	Market equity over book equity	Return on equity (ROE) against the cost of equity	Growth, payout, risk
EV/EBITDA	Enterprise value over pre-tax operating cash flow	Reinvestment rate (or return on capital)	Cost of capital, growth, tax rate
EV/Sales	Enterprise value over revenues	After-tax operating margin	Reinvestment rate, cost of capital, growth

Source: pptfiles/val3E/valpacket2spr25.md, slides “Price to Book Ratio: Determinants”, “EV to EBITDA - Determinants”, “The Determinants of EV/EBITDA” and “EV/Sales Ratio: Determinants” with notes (Spring 2025)

Three non-obvious lessons come with the formulas. First, PE moves more with growth when interest rates are low, because growth is a present value. Second, growth is not automatically good. A firm that grows while earning less than its cost of equity sees its PE fall as growth rises. Third, for PBV the rule is clean: an ROE above the cost of equity deserves a PBV above one, and below deserves below one. For EV/EBITDA his summary is: “Firms that deliver low growth, with high reinvestment and substandard returns on capital deserve to trade at low multiples of EBITDA.” For revenue multiples, “Operating margin dominates the regression in every market.”

Source: pptfiles/val3E/valpacket2spr25.md, slides “a. PE, Growth and Interest Rates”, “c. PE and Growth Quality”, “Price Book Value Ratio: Stable Growth Firm”, “The Determinants of EV/EBITDA” and “V. EV/Sales Regressions across markets” with notes (Spring 2025)

Why it matters for the calculator: the same inputs already sit in the model (growth, margin, sales to capital, cost of capital). A pricing panel can therefore show the multiple your own DCF inputs imply and set it beside the sector median.

Apply: comparables, mismatches, and sector regressions

A comparable firm is one with similar risk, growth and cash flows, not one in the same industry. “You can have firms in different businesses that have similar cashflow, growth and risk characteristics.” Since no two firms match exactly, you must control for differences. His four ways: direct comparison, story telling (this firm deserves a higher PE because it grows faster), a modified multiple such as PEG (PE divided by expected growth), or a regression when firms differ on several dimensions at once.

Source: pptfiles/val3E/valpacket2spr25.md, slides “Application Tests” and “The Control for Differences Choices” with notes (Spring 2025)

A mismatch is a company whose multiple does not sit where its companion variable says it should. His European bank example uses medians. A bank below the sector median PBV of 2.07, with ROE above the median of 11.82%, and volatility below the median of 21.93%, is a mismatch worth a closer look. The regression version for those 18 banks was $PBV = 2.27 + 3.63 \text{ ROE} \text{ minus } 2.68 \text{ standard deviation}$, with an R squared of 79%. Read it as: each extra 1% of ROE adds about 0.036 to PBV. The book applies the same matrix logic to revenue multiples set against margins.

Source: pptfiles/val3E/valpacket2spr25.md, slides “3: An Eyeballing Exercise”, “The median test” and “The Statistical Alternative” (Spring 2025); financeMD/Aswath Damodaran - Investment Valuation, University Edition _ Tools and Techniques for Determining the Value of Any Asset (2023).md, Chapter 20, “Looking for Mismatches”

A sector regression is a small statistical model that predicts a firm’s multiple from its fundamentals across a group. His market-wide PE regression on about 1,600 US firms in January 2025 predicted a PE of 33.75 for a firm with 15% growth, a beta of 0.90 (beta measures how much a stock moves with the market; 1.0 is average) and 20% payout, against a traded PE of 40.4. He then warns about the three usual problems: non-linearity, non-stationarity (the coefficients drift from year to year), and multicollinearity (growth and risk move together). Rules of thumb: a t statistic (how many standard errors a coefficient is away from zero) above 2 is good, 1 to 2 is marginal, below 1 is noise. Add one variable for every 10 to 15 observations. And a firm-level trap: Ryder showed a much lower EV/EBITDA than other truckers, most likely because it had the oldest fleet and a large fleet purchase ahead.

Source: pptfiles/val3E/valpacket2spr25.md, slides “Example 5: Overlooked fundamentals?”, “A Test on EBITDA”, “PE Ratio: Standard Regression for US stocks - January 2025”, “Problems with the regression methodology”, “Statistically insignificant?” and “Using the PE ratio regression” with notes (Spring 2025)

Why it matters for the calculator: a sector regression is a weekend project once the industry tables sit in Postgres, but it must carry its R squared and its date, because a low R squared means a wide prediction range.

When pricing beats valuing

Damodaran is honest about why pricing dominates. It is quicker, it needs fewer explicit assumptions, and “There is safety in numbers”. DCF ignores market mood, so it produces contrarian answers, and “if you are wrong, you will be wrong alone.” Pricing is the right tool when the goal is the price something will fetch today, as in an IPO. It also fits momentum strategies and managers who are judged against peers. The market also decides which denominator counts. In social media in 2013, enterprise value per user correlated with market cap at 0.98, better than revenues did. His instruction: “In pricing, it is the market that determines what matters.”

Source: pptfiles/val3E/valpacket2spr25.md, slides “Why relative valuation?”, “The Market Imperative”, “The Market sets the rules”, “Read the tea leaves” and “Statistically insignificant?” with notes (Spring 2025)

Why it matters for the calculator: the product values companies, but the memo always shows the price beside the value, and a pricing panel explains why the market disagrees with you.

Part B: What the whole market prices in

The implied ERP as a composite indicator

The equity risk premium (ERP) is the extra return investors demand for holding stocks instead of a riskfree bond. The implied ERP runs the valuation backwards. Take the index level as given, take expected cash flows (dividends plus buybacks) and expected growth, and solve for the discount rate that makes value equal to price. That rate is the expected return on stocks; subtract the Treasury bond rate and you have the ERP. The analogy is a bond’s yield to maturity: you know the price and the coupons, so you solve for the yield. Damodaran calls this number “my composite indicator for whether the market is richly priced or not” and stresses it “is entirely a market-driven number and is model-agnostic.”

Source: blog/2026/01/data-update-2-for-2026-equities-get.md (2026-01-23)

Here is the latest row he published, which your `erp_monthly` table mirrors column for column.

Item (start of September 2026)	Value
S&P 500 level	7686.14
US T.Bond rate	4.75%
Default-risk-adjusted dollar riskfree rate	4.53%
Expected growth rate (top-down, code TD)	13.99%
ERP (trailing 12 months, the headline)	4.09%
ERP with adjusted riskfree rate	4.31%
ERP with sustainable payout	4.14%
ERP smoothed (ten-year average cash flow)	6.05%
ERP normalized	3.56%
ERP net cash yield	3.80%
Expected return on stocks	8.84%

Source: pc/implprem/ERPbymonth.md, sheet “Historical ERP”, row 2026-09-01 (2026-09-01)

Two notes on that table. The adjusted riskfree rate exists because Moody’s downgraded the United States from Aaa to Aa1 on May 16, 2025. He nets out the 0.22% default spread for an Aa1 sovereign to get “a more consistent riskfree rate”. And the spread between the six ERP definitions, from 3.56% to 6.05%, is your honest error bar. The regime spec shows it as a band and calls the headline “soft” whenever the band is wider than 150 basis points.

Source: pc/implprem/ERPSept26.md, sheet “Impl premium calculator” (September 2026); claude/tool-results/btzaixzyh.txt, CHECK 3, SPEC 1 (2026)

The mechanics are simple. Base cash flow grows at the expected growth rate for five years, then at the riskfree rate forever. The discount rate is riskfree plus ERP. Excel goal-seeks the ERP until value equals the index. The September 2026 sheet solves to an implied expected return of 8.8437%, which is the 8.84% above (4.75% plus the 4.09% ERP). Its what-if row at an ERP of 4.25% gives an intrinsic index value of 7398.08 and an intrinsic trailing PE of 27.19. Those numbers are the golden test for `impliedErp.ts`.

Source: pc/implprem/ERPSept26.md, sheet “Impl premium calculator” (September 2026)

Why it matters for the calculator: the same ERP feeds every company valuation that month, so the regime panel and the company DCF read one stored vintage row.

The interpretive rule, in his words

He frames the reading as a question about the ERP, never as a forecast. If the market is “pricing in too low an ERP”, then you are contending that “stocks are over priced”. If the “current ERP is too high”, that equals arguing “stocks today are under priced”. And: “If you are not a market timer”, then “you are in effect arguing that the current ERP is, in fact, the right ERP for the market.”

Source: blog/2026/01/data-update-2-for-2026-equities-get.md (2026-01-23)

The fair-value grid

On January 1, 2026 the index stood at 6845.5, the expected return was 8.41%, and the ERP was 4.23% against a T.Bond rate of 4.18% (4.46% on the adjusted rate of 3.95%). To show how much the answer depends on the ERP you accept, he valued the index at ERPs from 2% to 6%.

ERP assumed	Index value	His label
2%	14834	“undervalued by 53%”
4.23% (the implied ERP)	6845.5	equal to the price
6%	4790	“overvaluation of 43%”

Then he declined to pick: “Rather than make that judgment for you”, he charted the implied ERP back to 1960. The current ERP is “almost exactly equal to the average for the 1960-2025 period”, lower than the post-2008 average, and far above the 2.05% of late 1999. The dataset behind that chart gives the averages: 4.25% for 1960 to 2025, 5.16% for 2006 to 2025, and 5.00% for 2016 to 2025.

Source: [blog/2026/01/data-update-2-for-2026-equities-get.md](#) (2026-01-23); [pc/datasets/histimpl.md](#), sheet “Historical Impl Premiums”, benchmark rows under the 2025 row

Why it matters for the calculator: the panel shows the ERP’s percentile against those three windows and reports a state (less risk priced than usual, near normal, more than usual), never a call.

Earnings yield against bond yield: context, not signal

Earnings yield is earnings divided by price, the PE turned upside down. It can sit next to the bond rate because both are yields. For 2025 the S&P 500 earnings yield was 3.97%, the T.Bond rate 4.18% and the T.Bill rate 3.65%. His 1960 to 2025 regression reads $E/P = 0.0341 + 0.5618 \text{ times the T.Bond rate minus } 0.1161 \text{ times the term spread (the T.Bond rate minus the T.Bill rate)}$, with an R squared of 47.4%. About half of the movement in earnings yield is explained by rates. Use this as background only. His verdict on PE as a market gauge is that it “is noisy and an unreliable indicator” and “would have suggested staying out of US equities for much of the last decade.”

Source: [pc/datasets/histimpl.md](#), row 2025; [pptfiles/val3E/valpacket2spr25.md](#), slides “A Counter”, “The Tie Breaker” and “Regression Results” (Spring 2025); [blog/2026/01/data-update-2-for-2026-equities-get.md](#) (2026-01-23)

His market-timing caveats, verbatim

- “I am not a market timer, but I do value the market at regular intervals”, and he does it “more to get a measure of what the market is pricing in, than to forecast future movements.”
- “I am a terrible market timer and try to avoid it in my investing”.
- “In bad news for market timers, none of the equity risk premium approaches does well at forecasting next year’s actual return”.
- “waiting on the sidelines for this to happen has not been a good strategy for the last decade”.
- His preferred outcome for 2026: “the healthiest development for the market would be for it to deliver a return roughly equal to its expected return (8-9%)”.

Source: [blog/2024/01/data-update-2-for-2024-stock-comeback.md](#) (2024-01-17); [blog/2026/03/the-price-of-risk-equity-risk-premium.md](#) (2026-03-15); [blog/2026/01/data-update-2-for-2026-equities-get.md](#) (2026-01-23)

Cadence matters for your cron jobs. He moved from an annual to a monthly ERP in September 2008. During crises (Brexit 2016, COVID 2020, tariffs 2025, the March 2026 oil shock) he estimates it daily.

Source: blog/2026/03/the-price-of-risk-equity-risk-premium.md (2026-03-15)

Why news sentiment never enters the computation

The implied ERP is computed from three things: the index level, trailing cash flows, and an earnings growth estimate. News is not one of them. Damodaran's reason is a doctrine, not a technicality: "markets are pricing mechanisms, not value mechanisms", and "Price is reactive, Value is proactive!" His 2026 opener is the proof. Someone who read only the news in 2025 "would have guessed" that stocks "had a bad year", and "You would have been wrong". The regime spec turns this into a rule. Sentiment may annotate the ERP chart with dated events and flag a divergence between news tone and the ERP. It may never be an input, and the panel may never be sorted by it. Store it in a separate `market_context_events` table and draw it as an overlay.

Source: blog/2020/03/a-viral-market-meltdown-iii-pricing-or.md (2020-03-16); blog/2026/01/data-update-2-for-2026-equities-get.md (2026-01-23); claude/tool-results/btzaixzyh.txt, CHECK 3, "What licensed news-sentiment APIs may legitimately add" (2026)

Why it matters for the calculator: the no-advice test and the "numbers go to the database, not free text" rule both depend on the ERP being a pure function of stored numbers.

Where this comes from

- pptfiles/val3E/valpacket2spr25.md: the whole relative valuation packet with speaker notes, slides 1 to 106. Read the notes, not just the bullets.
- New_Home_Page/lectures/multintr.md, pe.md, pbv.md, ps.md, vebitnote.md, relpe.md: short lecture notes, one per multiple.
- [financeMD/Aswath Damodaran - Investment Valuation, University Edition _ Tools and Techniques for Determining the Value of Any Asset \(2023\).md](https://financeMD/Aswath_Damodaran_-_Investment_Valuation,_University_Edition_-_Tools_and_Techniques_for_Determining_the_Value_of_Any_Asset_(2023).md): Chapters 17 to 20 (principles, earnings multiples, book value multiples, revenue multiples).
- blog/2026/01/data-update-2-for-2026-equities-get.md: the implied ERP, the fair-value grid, the PE critique, the "sidelines" line.
- pc/implprem/ERPbymonth.md and pc/implprem/ERPSept26.md: the monthly series and the live calculator with the goal-seek steps.
- pc/datasets/histimpl.md: annual implied ERP since 1960 with earnings yield, dividend yield and bond rates.
- blog/2024/01/data-update-2-for-2024-stock-comeback.md and blog/2026/03/the-price-of-risk-equity-risk-premium.md: the market-timing caveats and the forecasting evidence.
- blog/2020/03/a-viral-market-meltdown-iii-pricing-or.md: the price versus value doctrine.
- pdfs/country/val2dayIndia2025.md: the India session, with "Relative valuation or Pricing" defined and the line "Much of what passes for valuation is pricing."
- claude/tool-results/btzaixzyh.txt, CHECK 3: the regime panel spec built from all of the above.

Three things to remember

KEY IDEA

1. A multiple is a price with a ruler on it. Define it, describe its distribution, find its companion variable, then apply it only to firms with **similar fundamentals**.
2. **Pricing is not valuing**. Your calculator produces value; the pricing panel shows what the market pays, side by side, never blended.
3. The implied ERP (4.09% in September 2026, expected return 8.84%) tells you what the market is pricing in. Damodaran reads it as a state, refuses to time the market with it, and never lets news sentiment touch the arithmetic.

Chapter 7: India and the Rules of Thumb

KEY IDEA

Notice what changes when the country changes. Use the six steps and the rules-of-thumb page to keep currency, risk, tax and peer-table choices straight.

What this chapter is

Chapter 3 walked the DCF chain with Almarai, in Saudi riyals. Chapter 6 covered pricing. This chapter changes the country. Damodaran's claim is that the first principles do not change when you cross a border. What changes is a short list of inputs: the currency, the risk-free rate, the risk premium, the tax rate, and the peer tables. In his words, "the first principles of valuation are the same in all markets". Get those five inputs right and the rest of the chain runs unchanged. The chapter ends with his rules of thumb, gathered from across the corpus.

Source: [financeMD/damodaran/blog/2021/07/the-zomato-ipo-bet-on-big-markets-and.md](https://financeMD.com/damodaran/blog/2021/07/the-zomato-ipo-bet-on-big-markets-and.md) (2021-07-22)

Step 1: pick a currency and keep it

A currency is a unit of measurement, like meters or kilograms. It is not a source of risk. You can value an Indian company in rupees (INR) or in US dollars (USD). Both are correct if you are consistent, meaning every input is in the same currency: revenues, debt, cash, the risk-free rate, the growth rate, the share price, and the share count. The workbook has only two per-share cells, the price and the number of shares. Everything else is in whole-company currency units. The sheet warns explicitly about mixing millions with a raw share count. Indian filings add a second trap. They report in crore or lakh, and one crore equals ten million (a definition, not a corpus number). Convert once, at the door, and never again.

Source: [claude/tool-results/bkdrkpfng.txt](#), sections 6 and 7 (2026-09)

The currency choice also sets the terminal growth rate. The workbook's default is that growth in perpetuity equals the risk-free rate. In INR, that means the INR risk-free rate. Do not import a 2.5% dollar growth number into a rupee model. Do not import a 7% rupee growth number into a dollar model. The packet's rule for nominal cash flows is that "the growth rate should be nominal in the currency in which the valuation is denominated". For the calculator, this will become a single field, currency, stamped on every run, and a test will check that the growth cap uses the same currency's risk-free rate.

Source: financeMD/valpacket1spr25.md, slide 202 (Spring 2025); claude/tool-results/bkdrkpfng.txt, section 1.6 (2026-09)

Step 2: the INR risk-free rate

A risk-free rate is the return on an investment with no chance of default. A government bond only qualifies if the government is close to default-free. India is rated Baa3 by Moody's, which is not that. So the 10-year Indian government bond yield, the G-sec yield, contains a default spread. A default spread is the extra yield lenders charge for the chance of not being repaid. His fix is subtraction: INR risk-free rate equals the 10-year G-sec yield minus India's default spread. The reason is double counting. The cost of debt already adds the country default spread. If the risk-free rate also carried it, sovereign risk would be counted twice.

Source: financeMD/damodaran/pdfiles/country/val2dayIndia2025.md, slide 29 (2025); Valuation/docs/read-this-first.md, section 3.4 (2026-09)

His own India examples, all from the same seminar slide:

Case	Year	Rating and spread	G-sec yield	INR risk-free rate
Tata Motors	2010	Ba2, 3.00%	8.00%	5.00%
Vedant	2025	Baa3, 2.16%	6.32%	4.16%

In the Zomato posts he used 4.25% for July 2021 and 4.78% for July 2022, both after the same subtraction. When a currency has no trusted long bond, he uses inflation instead: the USD risk-free rate scaled by the ratio of expected local inflation to expected US inflation. His illustration, with a 4% dollar rate and 4% local inflation against 2.5% US inflation, gives 5.52%. That formula is also a sanity check on a G-sec yield you do not trust.

Source: financeMD/damodaran/pdfiles/country/val2dayIndia2025.md, slides 29 and 31 (2025); financeMD/damodaran/blog/2022/07/a-zomato-2022-update-value-pricing-and.md (2022-07-27)

Why it matters for the calculator: from milestone 4 the RBI 10-year G-sec yield will be hand-entered monthly as a market vintage row, and the engine will subtract the India default spread from the country risk vintage. Two vintages, one number, both ids stored on the run.

Source: plans/help-me-out-here-abstract-wilkinson.md, M4 (2026-09)

Step 3: the equity risk premium, mature plus country

The equity risk premium (ERP) is the extra return investors demand for holding stocks instead of the risk-free asset. For an Indian company, Damodaran builds it in two layers. Layer one is the mature-market ERP. Since the US lost its Aaa rating he computes it as the implied US ERP minus the Aa1 default spread. On 2026-07-01, with the S&P 500 at 7499.36, that was 4.42% minus 0.22%, giving 4.20%. Layer two is the country risk premium (CRP). He takes India's default spread and scales it up by the volatility of emerging-market stocks relative to emerging-market bonds, a ratio of 1.55 in July 2026. Total ERP for India is mature ERP plus CRP.

Source: financeMD/damodaran/blog/2026/07/country-risk-drivers-measures-and.md (2026-07-15)

The two 2026 vintages the calculator will store:

Vintage	Moody's rating	Default spread	Mature ERP	CRP	Total India ERP
January 2026	Baa3	1.87%	4.23%	2.85%	7.08%
July 2026	Baa3	1.75%	4.20%	2.72%	6.92%

The older seminar illustration uses the same logic with rounder numbers: a 2% default spread times 21 over 14, the Sensex volatility over Indian bond volatility, gives a 3% CRP.

Source: `claude/tool-results/bafb6h9rb.txt`, section 1 (2026-09);
`financeMD/damodaran/pdfiles/country/val2dayIndia2025.md`, slide 45 (2025)

One refinement matters for Indian names. Country risk comes from where a company operates, not where it is incorporated. His seminar pairs Tata Motors, with 91.37% of 2009 revenues in India, against TCS, with 7.62%. The workbook offers three ERP choices: country of incorporation, operating countries weighted by revenue, or operating regions weighted by revenue. Why it matters for the calculator: default to country of incorporation, but store revenue weights when the filing gives them, so an IT exporter is not charged full India risk.

Source: `financeMD/damodaran/pdfiles/country/val2dayIndia2025.md`, slide 52 (2025);
`claude/tool-results/bkdrkpfnfng.txt`, section 2.6 (2026-09)

Step 4: the tax rate

The workbook needs two tax rates. The effective rate is what the company actually paid last year, read from its own tax note. The marginal rate is the statutory rate on the next rupee of profit. The chain fades one into the other over years 6 to 10, as Chapter 3 explained. For India the marginal rate is a choice, not a lookup, because Indian companies can elect a regime:

Regime	Marginal rate
New regime (22% plus surcharge and cess; cess is an extra levy charged as a percentage of the tax itself)	25.17%
Old regime	34.94%
New manufacturing companies	17.16%

The country table inside the workbook shows 30% for India. That column is never read by any formula. It is information, not an input. Why it matters for the calculator: `marginal_tax_rate` will be a typed field with those three options, never copied from the country table.

Source: `claude/tool-results/bkdrkpfnfng.txt`, sections 3.2 and 6 (2026-09)

Step 5: industry averages and the fallback

The workbook ships two peer tables, US and Global, each with 94 industries. There is no India table inside it. Damodaran does publish India averages as separate files, 29 families over the same 94 industry names, built from 5,170 Indian companies. The catch is thinness. An average of three firms is not an average, it is three anecdotes. Counting from his beta file, 27 of the 94 Indian industries have fewer than 10 firms:

Firms	Industries
0	Utility (General)
1	Homebuilding, Oil/Gas (Integrated), Precious Metals, Reinsurance, Retail (REITs), Utility (Water)
2	Insurance (General), Semiconductor Equip
3	Insurance (Prop/Cas.), Retail (Building Supply), Transportation (Railroads)

Firms	Industries
4	Software (Entertainment), Telecom (Wireless)
5	Air Transport, Beverage (Soft), Coal & Related Energy, R.E.I.T.
6	Banks (Regional), Software (Internet)
7	Oil/Gas (Production and Exploration), Tobacco
8	Cable TV, Chemical (Diversified)
9	Drugs (Biotechnology), Electronics (Consumer & Office), Insurance (Life)

Why it matters for the calculator: the rule is India first, then emerging markets, then Global, switching whenever the row has fewer than 10 firms. The tier actually used will be written on every run as `industry_tier_used`, so a memo can say “peer margins are global, because only five Indian airlines exist”.

Source: `financeMD/damodaran/pc/datasets/betaIndia.md` (2026-01-05); `claude/tool-results/bafb6h9rb.txt`, sections 2 and 3 (2026-09); `Valuation/docs/read-this-first.md`, section 3.4 (2026-09)

Step 6: dollar-based averages and the inflation differential

Most of Damodaran’s industry averages are computed in US dollars. Ratios travel across currencies without change: a margin, a sales-to-capital ratio, or a beta is the same number in INR and USD. Rates do not travel. A cost of capital or a growth rate carries the inflation of its currency inside it. The workbook says so next to the price cell: “If you are not working in US dollars, you should add the inflation differential to the industry averages.”

Source: `financeMD/damodaran/pc/fcfsimpleginzu.md`, Input sheet note (valuation date 2026-02-01)

The workbook handles the cost of capital itself. Its industry-average and distribution approaches add the difference between your risk-free rate and 4.58%, the dollar risk-free rate of that vintage. The emerging-market distribution it offers is 7.63% at the first quartile, 8.95% at the median and 10.69% at the third quartile, all in dollars before that adjustment. His India file shows the full conversion for the whole market: 8.28% in dollars becomes 10.92% in rupees, using 5% expected INR inflation and 2.5% expected US inflation. The arithmetic is one plus the dollar rate, times one plus INR inflation, divided by one plus US inflation, minus one. Why it matters for the calculator: growth averages pulled from a dollar table will get the same treatment before they touch an INR run, and the rule will be logged. Margins pass through untouched.

Source: `claude/tool-results/bkdrkpfng.txt`, sections 2.6 and 6 (2026-09); `financeMD/damodaran/pc/datasets/wacclIndia.md` (2026-01-05)

His Indian cases, as published estimates

These are estimates he published, with the prices of the day. They show the method, nothing about the companies now.

Zomato, July 2021

Zomato is a food-delivery platform that listed in July 2021. His story: the Indian food-delivery market reaches about 25 billion dollars in ten years, Zomato holds about 40% of it, keeps 22% of order value as revenue, and its operating margin trends toward 30%. He assigned a 10% probability of failure and used a rupee cost of capital that carried India’s country risk. The value of equity came to about

394 billion INR, or 41 INR per share. The IPO price was 76 INR. He then replaced three point inputs with ranges in a simulation: market size from 10 to 40 billion dollars, share from 20% to 50%, margin from 15% to 45%. He noted that market size and share moved the value far more than the cost of capital did.

Source: [financeMD/damodaran/blog/2021/07/the-zomato-ipo-bet-on-big-markets-and.md](https://financeMD.com/damodaran/blog/2021/07/the-zomato-ipo-bet-on-big-markets-and.md) (2021-07-22)

Zomato, July 2022

A year later the price had peaked at 169 INR and fallen to 41.65 INR. He refused to call the fall a vindication, for three reasons worth copying. First, he also published a Paytm estimate that ended far above the price. Second, an intrinsic value should itself grow at the cost of equity, so 41 INR in 2021 becomes about 46 INR in 2022. Third, the company and the macro picture had changed, so it had to be revalued. India's ERP had risen from 6.85% to 9.08% and the rupee risk-free rate from 4.25% to 4.78%, adding roughly 1.5 to 2 percentage points to the cost of capital. The revalued estimate was 35.32 INR, down from 40.79 INR, most of the drop coming from macro inputs.

Source: [financeMD/damodaran/blog/2022/07/a-zomato-2022-update-value-pricing-and.md](https://financeMD.com/damodaran/blog/2022/07/a-zomato-2022-update-value-pricing-and.md) (2022-07-27)

Paytm, October 2021

Paytm is a mobile-payments platform. Its take rate, revenue as a share of payment volume, had fallen from 2.18% in 2016-17 to 0.79% in 2020-21. His story: the take rate stays low, reaching 1% in 2026, then doubles to 2% as the focus shifts from users to revenues. Operating margin reaches 5% in 2026 and 30% in stable growth. The rupee cost of capital was 10.43%, just below the median Indian company. The failure probability was 5%. The value of equity came to about 1,500 billion INR, close to 2,000 INR per share, against an unlisted price of 2,950 INR. His simulation showed a 3% chance the equity was worth nothing and a 90th percentile above 2,000 billion INR. By July 2022 the stock traded at 713 INR.

Source: [financeMD/damodaran/blog/2021/10/the-indian-smartphone-revolution-paytms.md](https://financeMD.com/damodaran/blog/2021/10/the-indian-smartphone-revolution-paytms.md) (2021-10-04); [financeMD/damodaran/blog/2022/07/a-zomato-2022-update-value-pricing-and.md](https://financeMD.com/damodaran/blog/2022/07/a-zomato-2022-update-value-pricing-and.md) (2022-07-27)

Why these matter for the calculator: they are the template for an Indian run. A story in five drivers, a rupee risk-free rate by subtraction, a country-loaded ERP, an explicit failure probability, and a range around the point. They also show why the accuracy scoreboard will store the estimate, the price and the vintages on the day, so later revisions are honest about what changed.

Source: plans/help-me-out-here-abstract-wilkinson.md, M4 (2026-09)

The 2025 India seminar themes

The two-day India seminar repeats the same frame with local cases. Its risk-free slide is the subtraction table above. Its country-risk slides argue that exposure comes from operations, not incorporation. Its stable-growth table is the clearest picture of a rule from Chapter 3: for Tata Motors, with return on capital equal to the cost of capital at 10.39%, the value stays at 435,686 million INR whether terminal growth is 0% or 5%. Growth adds nothing when the return on it only covers its cost. The seminar closes with the story-to-numbers steps, ending with "Keep the feedback loop open", and two warnings named "The Runaway Story" and "Willy Wonkitis", his labels for narratives that outrun any plausible market.

Source: [financeMD/damodaran/pdfiles/country/val2dayIndia2025.md](https://financeMD.com/damodaran/pdfiles/country/val2dayIndia2025.md), slides 29, 52, 60 and 149 onward (2025)

Rules of thumb, one page

KEY IDEA

Use this as your recap. Return to the earlier chapters for the reasoning behind each rule.

Each rule below is his, with where it comes from and what it becomes in the calculator.

Rule	Where he says it	In the calculator (planned)
Terminal growth never exceeds the risk-free rate, in the same currency. The risk-free rate is inflation plus a real rate; nominal GDP growth is inflation plus real growth; a mature firm cannot outgrow the economy forever.	valpacket1spr25.md, slides 202 and 203	A hard cap: $g_{\text{terminal}} \leq r_f$ in the run's currency. A run that breaks it fails.
Return on capital converges to the cost of capital in stable growth. Excess returns fade because competition arrives.	val2dayIndia2025.md, slide 60; workbook default assumption 2	Stable ROC defaults to the year-10 cost of capital. Overriding it is allowed but flagged.
A dollar cost of capital of 14% for a listed company is suspect. It sits above the 90th percentile for US firms. Eighty percent of US firms fall between 5.26% and 9.88%, global firms between 6.28% and 11.66%.	blog 2026/02, data update 5	A diagnostics warning when the dollar-equivalent cost of capital leaves the 10th to 90th percentile band.
Time spent finessing the cost of capital is mostly wasted. Cash-flow drivers move value more.	blog 2026/02, data update 5; Zomato 2021 post	Sensitivity tables rank inputs by effect on value, so the memo talks about the ones that matter.
Separate valuing from pricing. A DCF gives value. Multiples give price. Never blend them into one number.	Chapter 6; Zomato 2021 post, "The Pricing Game"	Two tables in every memo, never a weighted average of the two.
Keep the feedback loop open. Listen to people who know the business, and work out the value of alternative narratives.	val2dayIndia2025.md, slide 149	Every run keeps its story text, and a rerun with a changed story is stored next to the old one.
Do not trust a point estimate. Replace the three inputs you are least sure of with ranges and look at the distribution.	Zomato 2021 and Paytm 2021 posts	A Monte Carlo range (a simulation that draws each uncertain input from a range many times) built from his industry quartiles will ship with every valuation.
A currency is a unit. Pick one and stay consistent through the share count.	Model schema, section 6	One currency field per run and a consistency test on every input.
Country risk comes from where you operate, not where you are incorporated.	val2dayIndia2025.md, slide 52; blog 2026/07	Revenue-weighted ERP when weights exist; incorporation otherwise.
A big market is not a premium. The size is already inside growth and margin. The danger with big-market companies is an estimate that is too high, not too low.	Zomato 2021 post, "Big market delusion"	No premium fields exist. Market size enters only through revenue.
Your own scoreboard must record misses as loudly as hits.	Zomato 2022 post	The accuracy scoreboard will store every estimate against later prices.

Source: financeMD/valpacket1spr25.md (Spring 2025);
 financeMD/damodaran/pdfiles/country/val2dayIndia2025.md (2025);
 financeMD/damodaran/blog/2026/02/data-update-5-for-2026-risk-and-hurdle.md (2026-02-05);
 financeMD/damodaran/blog/2021/07/the-zomato-ipo-bet-on-big-markets-and.md (2021-07-22);
 financeMD/damodaran/blog/2022/07/a-zomato-2022-update-value-pricing-and.md (2022-07-27);
 claude/tool-results/bkdrkpfng.txt (2026-09)

Where this comes from

- financeMD/damodaran/pdfiles/country/val2dayIndia2025.md, the two-day India seminar; slides 29 to 31 for risk-free rates, 45 to 52 for country risk, 60 for stable growth, 149 onward for story to numbers.
- financeMD/damodaran/blog/2021/07/the-zomato-ipo-bet-on-big-markets-and.md, the full Indian worked case, including the simulation and the big-market delusion.
- financeMD/damodaran/blog/2022/07/a-zomato-2022-update-value-pricing-and.md, how to revise an estimate honestly when macro inputs move.
- financeMD/damodaran/blog/2021/10/the-indian-smartphone-revolution-paytms.md, take rates, a rupee cost of capital, and a wide Monte Carlo range.
- financeMD/damodaran/blog/2026/07/country-risk-drivers-measures-and.md, the 2026 method for mature ERP and CRP, and why operations matter more than incorporation.
- financeMD/damodaran/blog/2026/02/data-update-5-for-2026-risk-and-hurdle.md, the cost-of-capital distribution and the 14% test.
- financeMD/valpacket1spr25.md, slides 202 and 203, the growth cap and the risk-free rate as nominal growth.
- financeMD/damodaran/pc/datasets/betaIndia.md and wacclIndia.md, the India peer files, firm counts, and the rupee conversion.
- claude/tool-results/bkdrkpfng.txt, section 6, the cell-by-cell edit list for an Indian run.
- claude/tool-results/bafb6h9rb.txt, the dataset catalog with the India rows and vintages.

Three things to remember

KEY IDEA

1. **Five inputs change for India**, nothing else does: the currency, a risk-free rate found by subtracting the default spread from the G-sec yield, an ERP of mature plus country, a chosen marginal tax rate, and peer tables that fall back to emerging then Global below 10 firms.
2. **Rates carry inflation** and ratios do not. Convert dollar costs of capital and growth rates with the inflation differential before they enter a rupee run. Margins, betas and sales to capital pass through as they are.
3. His Indian cases are templates, not verdicts. Zomato at 41 INR against an IPO price of 76, Paytm near 2,000 INR against 2,950, both published with ranges, both revisited in print. Store the estimate, the price and the vintages, then let the scoreboard judge.

Source: recap of figures sourced in Steps 2, 5 and the Indian cases above (bafb6h9rb.txt; blog 2021/07 Zomato IPO; blog 2021/10 Paytm).